



WPI

Automation of Methodologies for Archival Processing

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Abstract

Under the guidance of Professor Marc Bui, we created methodologies to automate the archaeological classification process by developing pipelines that serve as the baseline for integrating artificial intelligence in database organization. This included developing methods for document preservation, textual analysis, and the construction of an authority file, as well as fine-tuning foundation models for image classification. These processes highlight potential implementations of AI tools for digital humanists.

Acknowledgments

The authors would like to thank our sponsor, Dr. Marc Bui, for his guidance through the design of our project. We are grateful to the participants who dedicated their time to this research in the form of interviews and informal advising sessions both in France and at WPI. Special thanks are extended to our project advisors, Professors Lindsay Davis and Gbetonmasse Somasse for their support through the duration of this project.

Executive Summary

Archaeology is grounded in interpretations of complex, fragmented historical data which takes the form of material remains; archaeologists spend their lives learning how to interpret datasets of highly-specific corpora. This is an innately time consuming process, and being able to extract meaning from archaeological data requires high levels of expertise (Daniel, 2026). Artificial Intelligence (AI) tools offer possible solutions to technical gaps in the field of archaeology, yet largely remain unexplored.



Figure 1: Digital scan of recorded rubbing of stele number 10628 (Vietnamica Gallery, n.d.).

Vietnamese steles, the archaeological corpus used in our project, are large, stone monuments engraved with text and stylistic motifs. The text contains records of the day-to-day life and religious, economic, and social practices of the seventeenth-century Vietnamese people (Scius-Bertrand et al., 2022). The motifs are highly stylized and they carry deep and complex symbolism. A record of these motifs as well as their characteristics and relationships can

be found in *L'Art a Hué*, a twentieth-century publication detailing the artistic practices of the Vietnamese city of Hué (Cadière, 1919). Improved access to information about these motifs, including multilingual translations in a digital format, would help streamline the classification process. This is the work of Professor Marc Bui, our sponsor, the Directeur d'Études Cumulants at École Pratique des Hautes Études.

The purpose of our project was to propose a generalized pipeline for archaeologists and historians to digitally preserve and analyze documents, enabling efficient and automatic classification of their contents.

Methodology and Findings

Our research was divided into two separate yet related research objectives. The first objective was creating a generalized pipeline for automated document preservation, processing, and textual analysis. The second was to explore the use of foundation models to detect motifs on a dataset and classify them based on entities extracted from the previously completed textual analysis.

The first objective was further separated into three sub-tasks: the creation of a digital document preservation pipeline, the creation of a textual analysis pipeline, and a method to organize the information from the document in a reference framework database. The document preservation pipeline is outlined in the following process:

1. Convert all files to the Tag Image File Format (TIFF) format using Adobe Photoshop.
2. Add metadata to the files using the Dublin Core (DC) standard using Adobe Bridge, which includes:
 - a. Generic metadata applicable to all files (parent document name, contributors, publication information, etc.).
 - b. File-specific metadata (file name, page number, contents, etc.).
3. Generate and apply a standardized file naming scheme.

The Adobe Creative Cloud Suite, which includes both Photoshop and Bridge, is widely recognized as industry standard software due to their robust capabilities and interconnectivity (Adobe, n.d.-b). The TIFF format is preferred by the United States Library of Congress for document preservation due to its lossless compression, meaning that no information is sacrificed when using this type of file (Adobe, n.d.-f). Dublin Core is a standard metadata format for all file types, and can be implemented with Adobe Bridge (Dublin Core Metadata Initiative (DCMI), n.d.).

We were successful in the execution of this pipeline, which resulted in all of the files being organized for long-term storage. Because the TIFF format was used, the pipeline produced files that retained all of their original quality and DPI information which allows for future document reconstruction. The metadata that we appended using Adobe Bridge and the DC standard enables easier organization in a document database, improving findability and utility by historians and archaeologists.

Textual analysis is the process by which a computer makes the text in a given file machine-readable (Amazon Web Services, n.d.-a). By performing textual analysis, the documents were transformed to fit the needs of the document preservation and database construction processes. The textual analysis pipeline is as follows:

1. Use ScanTailor to preprocess the files, which includes:
 - a. Fixing the DPI to the original of the scans.
 - b. Correcting the documents' orientation.
 - c. Deskewing the documents.
 - d. Selecting the documents' content (paragraphs, illustrations, annotations, etc.) and setting the margins.
 - e. Exporting as binarized (black and white) TIFF files.
2. Use Google Gemini's Optical Character Recognition (OCR) capabilities on the preprocessed files to produce a transcription and translation of the documents.

ScanTailor is a program that can perform all of the tasks outlined above with batch processing capability (ScanTailor, n.d.). The use of this tool was a recommendation of our sponsor. OCR is a frequently-used textual analysis method and can be easily performed by modern tools such as Large Language Models (LLMs) (Amazon Web Services, n.d.-a). Google Gemini, Google's LLM, had a flexible token limit that allowed for the processing of all files of interest, and far out-performed all other OCR tools that were tried. A standardized prompt was used for all files to ensure consistent outputs.

This pipeline also proved to be successful, and resulted in text documents containing all of the information from the original scans and a set of TIFF files ready for future image processing. With transcriptions and translations in an accessible format, we read and analyzed the documents to find key entities and relationships which both informed the metadata appended in the document preservation pipeline and contributed to the creation of the database.

The final sub-task in the first objective was the creation of a database of entities and relationships found in the text, also called an authority file or a reference framework. Authority files are controlled vocabularies that provide agreed upon terms, translations, and hierarchical relationships to be used in multidisciplinary and multilingual work (Filosa et al., 2023). The creation process was the following:

1. Read and analyze the transcriptions and translations of the reference text and manually identify all relevant entities and high-level relationships between them.
2. Add all identified entities into a spreadsheet that includes translations and relationships to other entities.
3. Create an object-oriented database in Neo4j by outlining Cypher queries that create nodes and relationships as described in the spreadsheet.

Spreadsheets, such as Microsoft Excel or Google Sheets, were used as they provide a convenient way to organize and manage information in a tabular format. Neo4j was selected as our database platform following interviews with database experts. They recommended object-oriented solutions such as Neo4j, as they allow for more unique entities and relationships compared to entity-relationship models (Ferguson and Hebels, 2003).

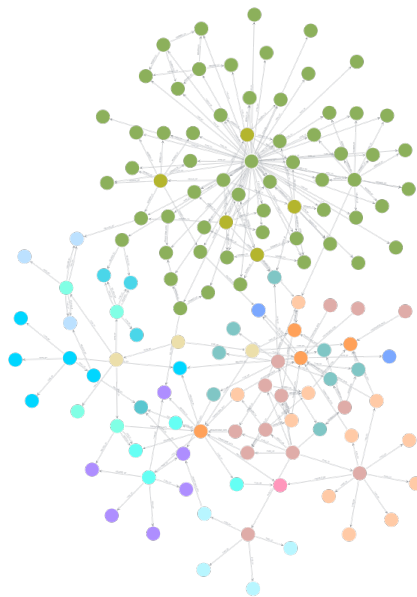


Figure 2: An image of the graphical representation of the authority file created by reading the English transcriptions of *L'Art à Hué*.

We were successful in creating a database of entities and relationships defined in our reference text, *L'Art à Hué*. Neo4j displays databases as a visual web of interconnected

nodes, so a visualization of the entities and the high-level relationships between them was also produced. With the database created, we could begin classifying the stele images using the terms that it contained.

Completing the first objective meant that we could start the second research objective, exploring the use of foundation models for automatic classification. We tried multiple tools, but we ultimately decided on a combination of Qwen-VL-8B for object detection and Meta's Segment Anything Model 2 (SAM2) for segmentation (Meta AI, n.d.; QwenTeam, n.d.). Detection is the process by which a computer analyzes an image, locating, identifying, and establishing a bounding box around any object of interest. Segmentation is a secondary step in which shaped masks are applied to the detected objects, determining exactly what pixels belong to the detected object of interest and what is undesired background or imagery (GeeksforGeeks, n.d.-b). To control for variation in the stylization of motifs, a set of two predetermined steles were used in all trials of detection and segmentation. We prompted the model to identify the objects as described in our authority file over a spectrum of box and text confidence threshold values, which are numbers between 0 and 1 that determine whether or not a correct prediction was made. The model's performance was then analyzed quantitatively using mean average precision (mAP), which averages the average precision (AP) scores across all object classes to provide an overall evaluation of accuracy (Anwar, 2022). Qwen proved advantageous over similar models such as GroundingDINO and Florence2, allowing for more robust prompting and parameter modification as well as having superior semantic reasoning. SAM2 is a widely-used image segmentation model developed by Meta. It was chosen as it has been widely used in pre-existing detection-segmentation pipelines and is easily installed and implemented locally.

Ultimately, all attempts were unsuccessful at automatic classification. The models drew bounding boxes around objects of interest to varying degrees of success, but none of them aligned well enough with the human-generated ground truth to be considered accurate classifications. This resulted in a mAP of 0.0000, which indicated no accuracy in classification whatsoever.

Conclusion and Recommendations

Our team was able to successfully research, develop, and execute pipelines for digital document preservation, processing, and analysis, as well as use the information extracted through these processes to create an object-oriented database for entity and relationship management. This database can be used as the basis for classification tasks, either manual or automatic. Although the results of the tested classification methods were unsatisfactory due to time,

cost, and technological constraints, the integration of AI tools in archaeological classification shows promising results and should continue to be explored to assist in this complex process.

Based on these findings, we propose the following recommendations for future development and iterations of this project:

Find or develop non-LLM solutions to textual analysis. Based on the ethical considerations discussed in Section 2.4, we determined that fine-tuning an existing OCR model or creating one from scratch capable of all three languages would be beneficial, as it could run locally and reduce the environmental impacts generated by prompting a large LLM like Google Gemini.

Explore means by which entity and relationship extraction can be automated. The current database methodology requires knowledge of Cypher, Neo4j’s query language, to complete the database (Neo4j, Inc., n.d.-a). Streamlining the process to an automated procedure would allow experts to implement the method without a coding knowledge requirement and would accelerate the completion of the database.

Upgrade existing detection-segmentation pipeline components and incorporate custom training data for detection and classification. Due to computational limitations, we were restricted in the size and subsequent performance of the models we were able to implement. By using larger, more powerful computers or paying to outsource computation, higher-performance models such as Qwen3-VL-235B-A22B can be integrated in the pipeline. In addition to this, exposing both the object detection and segmentation models to a variety of Vietnamese symbols and motifs would greatly improve their capabilities in identifying and classifying these types of objects. This exposition could take multiple forms, ranging from a small amount of reference images to a fully constructed dataset consisting of Vietnamese stele imagery.

Authorship

Kristen Thebeau, Sophia Perron, Franklin Weiss, and Thomas Ostrander all contributed to the research, writing, and editing of this report. A breakdown of how the report was written and the contributions to the findings is outlined below.

All

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- Case Studies (2.3.2)
- Methodology (3)
- Findings and Deliverables (4)
- Project Objective 1: Creating a General Pipeline for Archaeological Data Organization and Management (3.2)
- Project Objective 2: Creating an AI Pipeline for Image Recognition and Classification (3.3)

Kristen Thebeau

Kristen Thebeau was responsible for the code supporting the first project objective. This included the custom Dublin Core metadata template and the custom renaming script used in the "Method for Appending Metadata to Files for Document Preservation," as well as the construction of the authority file database in Neo4j. Ms. Thebeau was also responsible for the recorded documentation for the "Method for Appending Metadata to Files for Document Preservation" and the "Method for Creating an Authority File Database". Ms. Thebeau also formatted all of the documents in LaTeX.

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- Appendix D: Database Node Categories and Relationships
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- Appendix G: Outline of Metadata Field Contents
- Appendix H: Unique Field Metadata Outline
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- Appendix K: Entities and Relationships in *L’Art à Hué*

Sophia Perron

Sophia Perron contributed to this project by preparing research questions for expert interviews in French and English, as well as conducting interviews both online and in person in France. In addition to writing sections of the paper, Ms. Perron contributed to the preprocessing of various chapters of the reference text from thread one, as well as researching the ethical implications of AI use in archaeology.

- Acknowledgments
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- Ethical Considerations (2.4)
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- Appendix A: Sponsor Description
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Franklin Weiss

Franklin Weiss made all-around contributions to the completion of this project. Primarily, this took the form of developing the Optical Character Recognition pipeline and by doing the majority of the entity and relationship extraction to be used in the authority file. Mr. Weiss also appended all of the “Subject” metadata for the scans of *L’Art à Hué*.

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- Textual Analysis (2.2.2)

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- A Deep Learning Approach to Keyword Spotting in Vietnamese Stele Images (2.3.2)
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Thomas Ostrander

Thomas Ostrander assisted in the workload surrounding all aspects of the project. In addition to these contributions, Mr. Ostrander was primarily responsible for the development of the secondary thread, creation of the AI based object detection & segmentation pipeline.

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1 Introduction

Archaeologists dedicate their careers to studying the artifacts left behind by past civilizations and learning more about the history of humanity. Through processes of archival research, field work, analysis, and preservation, archaeologists gather data and answer research questions to further our understanding of the past (Society for American Archaeology, n.d.). According to Dwight Read, a Distinguished Professor Emeritus of Anthropology at the University of California Los Angeles, a critical part of the archaeological process is the accurate classification of uncovered artifacts. These classifications may be based on a number of features or assumed origins, but “should make evident the cultural context that gave rise to the patterning expressed through a classification” (Read, 2018). A logical classification system allows archaeologists to recognize patterns within the dataset and search through large datasets with ease. However, as new discoveries increase the size of these datasets, the time archaeologists must spend in the classification process only grows. Exploring new approaches to these processes is therefore becoming increasingly important; and the rise of new automation technology may prove to be the solution.

Automation is becoming increasingly integrated into modern workflows. While many are concerned with the way in which automation is reshaping the workforce, others are benefiting from the advancements that simplify their most complex and routine tasks (Subha Nandhini et al., 2026). As we will discuss, recent studies have investigated the potential integration of artificial intelligence in archaeological classification systems, and many have made advancements towards successful models. Marc Bui, a Professor of Computer Science at the University of Paris 8 and the Director of Studies at the École Pratique des Hautes Études, has contributed to these advancements with his work regarding Vietnamese steles. Vietnamese steles are large, engraved stone monuments erected during the restored Le dynasty. They record information on the day-to-day life of Le dynasty villagers in the mostly-obsolete Chu Nom script, constructed from a combination of Chinese characters and characters unique to the script itself (Scius-Bertrand et al., 2022). Professor Bui and his team investigated the application of artificial intelligence for automatic keyword spotting in the textual content of the Vietnamese steles, without the need for preprocessing, segmentation, or human annotation (Scius-Bertrand et al., 2022). This allows for the retrieval of steles that contain a certain search term and eases the search process in large databases of steles. After successfully defining this pipeline, Professor Bui is now investigating similar identification methods for the motifs depicted on the margins of the steles.

In order to define an automated system a standardized process and terminology must be defined. This research was completed in two stages. The first stage began with standardizing

and automating document preservation and textual analysis processes in order to build a database of reference terms used to classify the motifs in the margins of the steles. The reference terms were collected from the *L'Art à Hué* artbook, which contains a collection of art, architecture, and decorative motifs of Hué, the former capital of Vietnam during the Nguyễn Dynasty in the nineteenth century (Paris Musées, n.d.-b). With a set of defined reference terminology, the investigation moved to the second stage, with an exploration of artificial intelligence applications for object detection and semantic segmentation on a dataset containing scans of seventeenth century Vietnamese stele rubbings.

The remainder of the paper is structured as follows: the background chapter discusses the historical and cultural context of *L'Art à Hué* and Vietnamese steles, archival processes, previous studies and background in the applications of artificial intelligence in archaeology, and ethical considerations observed throughout the project. The methods chapter outlines the methodology for each stage of the project, while the findings chapter outlines the resulting procedures for document preservation, textual analysis, and the creation of a reference framework, along with a potential implementation for image classification on the Vietnamese stele dataset.

2 Background: Archaeological Context, Archival Processing, and Artificial Intelligence

This chapter outlines information relevant to the execution of our project. This includes the archaeological and cultural context, details on archival processes, and a brief explanation of the implemented artificial intelligence tools.

2.1 Archaeological and Cultural Context

Archaeology is the science of studying human-made artifacts with the ultimate goal of enriching our present body of knowledge with the advancements made by past civilizations. These artifacts range from the most primitive of Stone Age tools to the imposing cathedrals of Christendom Europe. Archaeology is a “principal source of knowledge of prehistoric, ancient, and extinct culture,” but this source comes at the cost of laborious work in the description, analysis, and classification of artifacts (Daniel, 2026). Many archaeologists spend their entire careers developing a more objective taxonomy to better place recovered artifacts in their proper historical contexts.

Technology has long aided in archaeological exploration, such as post-war aerial photography and electric and magnetic prospecting (Daniel, 2026). Many archaeologists and computer

scientists agree that artificial intelligence (AI) will be the next resource in the modern archaeologist’s kit, enabling more accurate and efficient work. Because of the high-level of expertise required, multiple studies have been conducted exploring the possibility of using AI in archaeology, especially to aid in the task of classification (see Section 2.3.2).

2.1.1 *L’Art à Hué*

L’Art à Hué is a book which studies the art, architecture, and decorative motifs of Hué, the former capital of Vietnam during the Nguyễn Dynasty, which took place from 1802 to 1945. The book was created by a group of French scholars (L’Association des Amis du Vieux Hué) dedicated to cultural heritage studies (Paris Musées, n.d.-b). The primary researcher was Léopold Michel Cadière, a highly respected Catholic missionary and linguist who specialized in Vietnamese history and culture. The book is a unique collaboration between French scholars and Vietnamese royal artists, as imperial artists of the Nguyễn court directly contributed to the work.

The book was created as a reference text to catalog and analyze the intricate art of Hué; it contains seven major categories of motifs: Geometric Ornaments, Characters (writing), Inanimate Objects, Plants, Animals, Landscapes, and Sculpture. The book was published originally as an essay in 1919 as an issue of the *Bulletin des Amis du Vieux Hué* magazine; this is the edition of the text we used for our research. Later editions of the book were published between 1920 and 1930. The original text was written only in French, taking an approach intended for academic audiences. This historically limited its accessibility to the general public, until later Vietnamese translations were published. This text served as our reference text to inform the vocabulary in the authority file applied to the annotations of the Vietnamese stele dataset.

Our team was provided with a set of scans, in the form of PNG files, of the original text by Professor Bui. The scans were used as the subject of the document preservation and textual analysis processes, as well as the source of the reference vocabulary defined in the authority file that were used to label the motifs on the dataset of Vietnamese steles.

2.1.2 Vietnamese Steles

Many of the motifs defined in *L’Art à Hué* appear on steles, or large, engraved stone monuments which were erected during the restored Le dynasty. They record information on the day-to-day life of Le dynasty villagers, including records of religious, economic, and social life (Scius-Bertrand et al., 2022). They were engraved in the mostly-obsolete Chu Nom script, which uses both Chinese characters and ones unique to the script. Due to its hybrid

nature, the Chu Nom is very complex and inaccessible. It was widely phased out in favor of the modern Vietnamese script introduced by Catholic missionaries in the seventeenth century, and there is a limited set of experts, globally, capable of translating the text today (Li, 2019).

Effects of weathering and deterioration only increase the challenge of translation and comprehension, making the successful reading and subsequent analysis and classification costly and time consuming. A digitized dataset of the stele images was generated by stamping the pattern onto a sheet of paper and then taking a digital photo of the stamped image. The digitized images often provide better legibility than the original steles, and their digital nature allows for further computer processing and analysis. However, the lack of access to experts still proves a monumental obstacle in the extraction and classification of the unique history contained in the steles, which gives rise to the exploration of an automated pipeline for this classification (Scius-Bertrand et al., 2023).

2.2 Archival Processes

This section details the importance of document preservation, textual analysis, and authority files when working in the digital humanities.

2.2.1 Document Preservation

The preservation of digital and physical documents allows for the continuous storage and access of records and files. The goal of digital preservation is to accurately render authenticated content over time to ensure its authenticity, accessibility, and usability (United Nations, n.d.). The life span of physical documents has become less concerning in the last century, as scientists have found methods to preserve paper for up to 1,000 years (Castillo, 2009). However, the required space to store large amounts of physical documents is unrealistic. To reconcile this, historians have turned towards microfilm and digital storage solutions. The digitization of these records encounters a similar issue, however, as the amount of digital storage required for large corpuses is enormous and expensive. Technicians have compiled methods for file formats,



Figure 3: Digital scan of recorded rubbing of stele numbered 10628 (Vietnamica Gallery, n.d.).

metadata, and other recommendations that allow for lossless compression techniques that reduce storage requirements while maintaining file quality. The Library of Congress, in their 2025-2026 Recommended Formats Statement, lists several recommendations for metadata and file format in digital libraries. The figure below outlines their recommendations for image files, which are applicable to the scans of *L'Art à Hué* and to the dataset of Vietnamese stele images (Library of Congress, n.d.-b).

ii. Photographs – Digital		
	Preferred	Acceptable
A. Faithful representation of the work	- Equal in quality to the published version, best edition or master copy - In the same format as the master copy	
B. Technical Characteristics	- Highest resolution available, not rescaled or interpolated - Highest bit depth available, 16 bits per channel if available - Embedded color profile or specified color space used in published version - Uncompressed - Unlayered	- Lossless compression or lower compression ratios - Discrete wavelet transform (DWT) preferred to discrete cosine transform (DCT) - Layered, if supported by preferred or acceptable format
C. Formats	TIFF (.tif) JPEG2000 (.jp2) PNG (.png) JPEG/JFIF (.jpg)	Photoshop (.psd, .psb) JPEG2000 Part 2 (.jpf, .jpx) Digital Negative DNG (.dng) - Proprietary Camera Raw formats (.nef, .crw, .arw, .iiq) GIF (.gif)
D. Metadata	- As supported by format: - Title - Creator - Creation Date - Place of publication - Publisher/producer/distributor - Contact information - Include if available: - Common embedded schema (e.g., IPTC) - Language of work - Other relevant identifiers (e.g., DOI, LCCN, etc.) - Subject descriptors - Abstracts - Key or reference to each data field and technical production information (e.g. EXIF metadata from digital camera)	Metadata provided separately in external text, CSV or XML -based sidecar file

Figure 4: Library of Congress Recommended Formats Statement for 2025-2026 (Library of Congress, n.d.-b). This table outlines the preferred storage and metadata information for digital photographs. The *L'Art à Hué* artbook was stored as a series of single-page PNG files, so these recommendations were the most applicable.

The Dublin Core Metadata Initiative (DCMI) is a compilation of metadata specifications and terms that serve as the international standard for metadata. There are 15 elements in the original Dublin Core publication that were adopted in 2001 as the United States of America standard and in 2003 as the international standard (Dublin Core Metadata Initiative (DCMI), n.d.). The 15 elements are contributor, coverage, creator, date, description, format, identifier, language, publisher, relation, rights, source, subject, title, and type (Adobe, n.d.-d). Using widely-accepted standards for data storage and metadata applications enabled a generalized understanding of methods that may be passed between researchers with limited need for clarification.

2.2.2 Textual Analysis

Textual analysis is the process by which a computer makes the text in a given file machine-readable. Image files, like TIFFs, are preferred for document preservation as they retain all of the information from the source; however, the text contained within the images cannot be read by computers, which makes tasks such as translation, keyword searching, and other manipulations impossible without an intermediary step. Optical Character Recognition, or OCR, is one means by which this can be accomplished. OCR scans a document and classifies the dark areas as text, or glyphs, and the light areas as background. It then compares the glyphs it identified to a large set of reference glyphs and assigns characters based on the closest matches (Society for American Archaeology, n.d.). By performing OCR on our reference document, *L' Art à Hué*, we will be able to create accurate translations using tools like Google Gemini, and the translations can be read and analyzed to find the terms for our authority file.

2.2.3 Authority File

A crucial component of effective image classification is the construction of a sufficient authority file. An authority file, or reference framework, is a standardized vocabulary that prevents inconsistent or ambiguous classifications, which is especially useful when operating in multidisciplinary contexts such as digital humanities (Filosa et al., 2023). For applications in automation and, specifically, artificial intelligence, the authority file acts as the basis for classifications by serving as the defined list of possible labels for each entity.

2.3 Artificial Intelligence

This section introduces foundation models, their relevance to our project, and examines previous attempts at integrating artificial intelligence technology in the field of archaeology.

2.3.1 Foundation Models

Foundation models are a type of artificial intelligence (AI) model, capable of fulfilling a broad range of general tasks. They are generally more applicable than typical machine learning models, especially when considering the specific implementation and prompting used for these models. Foundation models are described as AI models that have undergone training on large amounts of data before release, which gives the model the capability to excel in a specific task or field depending on its training data (R. D. C. Stryker, 2025). Large language Models (LLMs) are models that have been trained on large amounts of text and code. This

includes popular models such as OpenAI’s ChatGPT, DeepSeek, Anthropic’s Claude, and many others. Other foundation models, such as Alibaba’s Qwen variants, are more applicable to our specific use case as they are trained with an emphasis on imagery.

Models that have undergone image specific training, such as the Qwen variants previously mentioned, are referred to as Vision Language Models (VLMs). These models are another type of foundation model that appears adjacent to the better known LLMs, able to be interacted with through a chat interface or implemented through code for more developer specific use cases. Foundation models are able to take input prompts from a user, instructing the model on the desired process and output to better tailor the model to the specific task at hand. The technology introduced above was the basis of our exploration into artificial intelligence models capable of automating the archaeological image classification process.

2.3.2 Case Studies

To better understand previous efforts made to integrate artificial intelligence in the field of archaeology, an investigation of past studies in the intersection of artificial intelligence and archaeology was conducted. The sections below contain a summary of the findings of each study and how they are applicable to the AI classification in the dataset of Vietnamese steles.

Automated Identification and Classification of Rock Art Images

Rock art carvings are images on exposed bedrock created by using percussive force to remove parts of the rock surface. A team at the University of Gothenburg sought to automate the identification and classification of these carvings in order to reduce archaeologists’ time and effort spent with any one image (Horn et al., 2021). The challenge they encountered is similar to that of the Vietnamese stele dataset; how to train the model to recognize the varying symbols of each motif on an artifact at any scale, position, and orientation. The main difference between the two studies is the data structure. Their data was constructed of three-dimensional scans of rock art carvings, whereas the Vietnamese stele dataset is composed of two-dimensional rubbing scans.

The team used an approach that applied a CNN-based algorithm for object detection. Object detection is “a [computer vision] technique that uses neural networks to localize and classify objects in an image” (Kavlakoglu, 2024). After the team generated images from a 3D mesh, they implemented a faster region-based convolutional neural network (faster R-CNN). A faster R-CNN is an object detection model that identifies objects in an image and classifies them with bounding boxes (Singh, 2024). The architecture the team employed involved three sub-networks. The first was a ResNet-50 network for feature extraction. In this network,

the features contained in the input images were extracted into a feature map. The feature map is then fed into a region proposal network (RPN) to propose regions of the image that most likely contain objects. The RPN contained three layers; a convolutional layer that then fed into two sibling convolutional layers. One was a box regression with a linear activation function, while the other was a box classification with a sigmoid activation function. The final network was a classification network that proposed the bounds of the regions in region of interest (RoI) pooling.

a	No Augmentation No Transfer Learning	With Augmentation No Transfer Learning	With Augmentation With Transfer Learning
Animal	0.0	4.8	8.8
Boat	4.7	42.0	58.0
Circle	0.0	7.1	31.4
Human	0.0	10.2	25.3
Wheel	0.0	0.0	21.6
mAP	1.0	12.9	29.0

b	Image Type 1 Curvature Removal	Image Type 2 Transformation	Image Type 3 + Blending with Normal Map
Animal	5.6	8.8	7.0
Boat	40.1	58.0	51.3
Circle	57.1	31.4	22.2
Human	27.7	25.3	21.5
Wheel	21.6	21.6	15.9
mAP	30.4	29.0	23.6

c	Multi Type 2 (Type 1 + Type 2)	Multi Type 3 (Type 1 + Type 2 + Type 3)
Animal	7.1	10.0
Boat	64.1	61.3
Circle	40.0	33.3
Human	28.9	24.0
Wheel	22.2	18.8
mAP	32.5	29.5

Figure 5: Results of the University of Gothenburg faster R-CNN model. The values are representations of the mean average percentage (mAP), scaled up by 100. Table (a) is a model without data augmentation and transfer learning; table (b) is a model with different input images; table (c) is a model with multiple image inputs. (Horn et al., 2021).

The main problem archaeological classification models face is overfitting due to the amount of and types of data available. Overfitting occurs when a model fits too closely to its training data, causing inaccurate predictions for data other than the training data (“Overfitting,” n.d.). This is mainly caused by a lack of variability in the training data, meaning the model recognizes only a few objects and attempts to classify everything within this scope. This team attempted to reduce overfitting by fabricating additional data, training on several rotations, and tiling the images into multiple sub-images with specified overlap, which allowed the

model to classify the same object twice on different images. The results in terms of mean average precision (mAP), are shown below. mAP combines the model’s precision and recall to produce a measurement of its classification accuracy (GeeksforGeeks, n.d.-a). The scores range from 0 to 1 and represent a percentage metric of correctly classified objects and recalled objects, so a higher mAP score means the model performed better. The results in Figure A prove that the overfitting reduction methods the team at the University of Gothenburg used in their faster R-CNN model were successful, as the mAP values in the augmentation and transfer learning column are consistently higher than those without.

A Deep Learning Approach to Keyword Spotting in Vietnamese Stele Images

As mentioned previously, Vietnamese steles are large stone monuments engraved with information on economic, social, and religious practices from sixteenth to twentieth centuries Vietnam. This engraving was often done in the Chu Nom script, which is difficult for modern archaeologists to read and understand correctly. As a potential solution to the difficulty in identifying Chu Nom characters, a combined team of researchers, including Professor Marc Bui, from University of Fribourg and HES-SO in Switzerland and Ecole Pratique des Hautes Etudes in France proposed using a hybrid machine learning model to automatically classify extracted text on these steles. They used the YOLOv5 algorithm, trained on synthetic stele images, to detect characters in digital images of the steles (Scius-Bertrand et al., 2023). YOLOv5, an acronym for “You Only Look Once,” is a computer vision model developed by Ultralytics that prioritizes speed and accuracy. Unlike other models, it “processes the entire image in one go, making it significantly faster compared to the region-based approach of RCNN, which involves multiple passes” (Jocher, 2023). Then, processed the images further, creating graphs with vertices at key points (intersections, endpoints, and circular structures) as well as periodically placed at a fixed interval. These graphs were then used by Edit Distance methods, which calculate the theoretical minimum number of edits to transform one graph into another, to discern similarity between characters.

The research team found that when searching for a specific keyword character, a mean average precision (mAP) of 70% could be achieved when the style of the keyword matched those of the stele characters. However, that figure dropped to around 40% when the styles did not match, indicating more improvements to be made to the methodology, the model, or both. Suggested further research and work includes style clustering, noise reduction/removal, and the use of data-driven approaches instead of synthetic training sets. Ultimately, the team hoped to be able to generalize this technology to many different scripts and languages to be able to better learn from the writings of past civilizations (Scius-Bertrand et al., 2023).

Neural Network for Archaeological Glyph Detection

This study investigated the use of Convolutional Neural Network (CNNs) to detect ancient glyphs, a type of Neolithic rock art on the interior of caves. The researchers propose a neural network-based approach to incorporating a dedicated preprocessing phase for the data used to train the model. Their methods focus on quantifying the depth of the engraving in the image pixel, so that a defined variable (PIXEL) is a function that assigns a real value to each pixel. This variable then represents the depth of the engraving across the entire image grid (Crisci et al., 2025). After this variable is defined, the identification rule is based on the distribution of a statistical variable. This is a key step in isolating objects of interest from the background of the image.

The preprocessing technique in this paper is notable for the classification of Vietnamese steles. It serves the purpose of making the glyphs stand out from the background before they are processed by a CNN to classify them. It does so by converting the image to grayscale and then using a statistical technique (Turkey’s method for outlier detection) to identify pickles that are statistical “outliers” (meaning darker than the rock background) (Crisci et al., 2025). This technique was used for glyphs because the carvings in the rock make dark shadows, often not distinct shapes, that are unnatural to the rock surface. This process makes the glyph separable from the background, even in damaged, poorly lit images (Crisci et al., 2025).

The authors test 3 existing CNN models: AlexNet, GoogLeNet, and EffcientNet-B0. Each model was trained on a large curated synthetic datasets before being tested. The researchers found that the preprocessing step improved classification accuracy. GoogLeNet was the most accurate model, achieving 98.7% accuracy, after being trained on synthetic data and tested with real images (Crisci et al., 2025). The researchers concluded that their preprocessing technique in combination with a CNN is an accurate and effective tool for archaeology. GoogLeNet is recommended as the best model for practical use. They consider this use of AI as a non-invasive digital heritage tool that supports digital heritage initiatives.

AI-Assisted Digitalisation of Historical Documents

The preservation of archaeological artifacts often involves their digitalisation, or the process in which their information is uploaded to a database in a variety of manners ranging from 3D scans to simple pictures. For archaeological objects such as manuscripts, this process is not adequate. The digitalisation of their content is required in order to fully understand their context and information. Digitalisation involves the transcription of scanned texts, which more often than not involves an in depth and laborious process of deciphering said text. The goal of this study, *AI-Assisted Digitalisation of Historical Documents*, was to analyze the

efficacy of supplementing the work of experts with artificial intelligence, more specifically Deep Neural Networks (DNNs) to decipher text (Ferro et al., n.d.).

The study specifically aimed to determine the number of handwritten lines of historical documents that a DNN requires to be properly trained for Handwriting Text Recognition (HTR). The study explored the idea of involving data augmentation and transfer learning to these models as a means of achieving a lower character error rate (CER) with a limited dataset. Data augmentation involves the modification of existing data with known techniques such as affine transformations or rotations and translations (Ferro et al., n.d.). These methods alter the data in a coherent manner, resembling variations that are realistically encounterable. Transfer Learning involves the use of a pre-trained model on a distinct but similar dataset, then utilizing this model as a starting point for the model to learn a new one (Ferro et al., n.d.).

The datasets used for the pre-training of the network consist of images of lines of text, specifically the IAM dataset which is a modern English language dataset containing roughly 1500 pages of text from 657 authors and more than 13,000 scanned lines of text (Ferro et al., n.d.). The datasets used for benchmarking of the model or the ‘target’ datasets are those commonly used in historical documents, namely the Saint Gall, Parzival, and Washington datasets. These ‘target’ datasets were then subjected to varying experiments used to determine the CER as a function of the number of training samples over the following cases: baseline case with no modification, dataset with data augmentation and no pre-training, and the dataset with both data augmentation and pre-training. The results of these experiments display that the inclusion of data augmentation improves the CER across the board, however the use of pre-training can increase the error rate due to differences in the source and target dataset as well as the insufficient amount of target data than can be leveraged to adjust the model properly.

Authors of this study conclude that the use of this AI-based method is sufficient for supplementing the efforts of experts in the transcription of historical documents. At its current state the application of data augmentation techniques to the ‘target’ dataset and fine-tuning using modern handwriting as the ‘source’ dataset can be effective in historical handwriting recognition when the volume of training data is limited (Ferro et al., n.d.). The recognition error rate at the current state can be regarded as insufficient and as such this method should only be considered as an aid to experts. This is representative of the data collected regarding the general scope and application of the model for Vietnamese steles. In conferring with experts in archaeology and history, it can be concluded that academics in applicable fields are open to the addition of artificial intelligence to their work and view AI as a supporting tool that has little chance of replacing professionals in the field.

2.4 Ethical Considerations

Process augmentation with AI is not a new concept to the field of the digital humanities. Critics argue that the integration of AI into archaeology is inherently reductionist. The reliance on oversimplification of complex, interpretable archaeological data for computational ease risks a deterministic viewpoint (Gattiglia, 2025). Classifications based on definitive truths make much room for bias. A philosophical line of questioning is worth exploring when discussing the ethical limitations of AI integration into archaeological image classification.

Archaeology is about understanding human history through material remains. Extracting insights from archaeological findings about human culture, history, and nature requires a phenomenological approach. Phenomenology in archaeology refers to the principle that all humans have generally similar spatial perceptions of body centric experiences (Dyke and M, n.d.). To understand humans of the past, previous landscapes must be explored through this lens. A conceptual approach is necessary to personify archaeological findings.

Vietnamese steles are representations of a person’s life; simplifying them to material remains removes the phenomenological perspective necessary to conceptualize intended meaning. In the case of Vietnamese steles, a holistic interpretation requires an understanding of the text on the stele as well as the symbolism behind the imagery, as well a base understanding of the cultural context in which the artifact existed.

The object of this project was not to make conclusions about the content on Vietnamese steles. Our goal instead was to investigate computational methods to identify individual motifs found on steles. This objective existed in conjunction with the other thread of our research, focused on digital document preservation and analysis, to attempt at augmenting current archival processes with AI.

Our project involved the use of Large Language Models (LLMs). It is important to explore potential bias when using an LLM. Many of the existing LLMs are trained on large datasets “based on the wide knowledge of the internet,” (Tenzer et al., 2024). This is concerning because using an outsourced pretrained model means that there is an embedded bias of creation (western-centric colonial). It is also important to recognize that pre-existing LLMs rely on industrial infrastructure (data centers, rare minerals, cheap labor), with a massive carbon footprint (Gattiglia, 2025). Computation can quickly become energetically expensive; the environmental and labor costs are directly linked to using these models and therefore must be considered.

It is also important to reflect on the broader social considerations of using a LLM owned by a large tech monopoly. At different parts of this project we used Google Gemini, Deepseek, OpenAI’s ChatGPT, Qwen, and Microsoft Copilot. Via the nature of these AI models, using

them ultimately contributes to training them. Using them in casual settings also contributes to a culture of normalization around the support of these resources. This is sensitive when we consider that multiple of these companies (Google, OpenAI, Microsoft), have multi million dollar contracts with the United States Department of War (U.S. Department of War, n.d.). This makes using these resources inherently politically charged. This consideration only scratches the surface of the discussion on AI use and ethics. Ethically unsound applications of these AI models are being further developed daily (ex. mass surveillance, predictive policing, use of bias systems in healthcare). We do not aim to contribute to the normalization of casual AI use due to its unavoidable support of unsustainable practices.

3 Methodology

The research objectives outlined below build upon one another in support of the overall goal of this project: establishing a generalizable process for the automation of database organization and image annotation and classification.

1. Identify methods of computer-based document preservation and textual analysis to create accurate digital representations for reconstruction and further processing. This also involves the creation of a text-specific reference framework (authority file) and a visual conceptual map that illustrates hierarchical relationships between concepts from the document(s).
2. Explore and evaluate an approach for the classification of motifs on images of Vietnamese stele rubbings using artificial intelligence. The classification labels should be based on the agreed-upon vocabulary defined in the authority file in Objective (1).

Through the proposed methodology, this project contributed to larger discussions on the applications of AI for the digitisation of historical works, ideally making it a more accessible and useful tool to digital humanists. The chapter below describes the implementation methods used to execute these objectives.

3.1 Expert Interviews

To inform our methodology, we conducted interviews with seven total experts from the fields of computer science and archaeology, as well as history and religious studies. We did so to get a better understanding not only of the complex history behind steles, but also to contextualize our technical goals within the current research of experts in the digital humanities. All of

the interviews were conducted in an informal, semi structured manner. We used snowball sampling to locate better fit experts as the interview process went on.

Four of our seven interviewees yielded from WPI's Computer Science department. We chose specialists in computer vision, database engineering, as well as artificial intelligence to provide insights on what existing tools we should explore. We prepared a mix of both general questions about existing options and niche questions about our particular object of study. Two of the interviews took place before the start of our project. These were more general, and functioned to introduce us to possible directions for our research objectives. The two that took place after the start of our project took on a more structured format, by that time we had specific technical questions and needed more direct guidance on our defined object of study.

In preparing for our interviews with the humanities experts, we researched the general nature of steles as well as the historical context in which they exist. We did so from a religious perspective as well; a lot of the imagery on the steles is reflective of Buddhist and Hindu ideologies, so we familiarized ourselves with relevant concepts. We developed around twenty research questions in French and English. In aim to better understand the integrated nature of the imagery on the steles the humanities interviews took on a conversational structure. Due to our lack of expertise within this context it was best to allow the interviewee to indicate the direction of the discussion, we adapted our line of questioning in real time based on responses.

3.2 Project Objective 1: Creating a General Pipeline for Archaeological Data Organization and Management

The first project objective was to create a general pipeline for archaeological data organization and management. This objective consisted of three sub-stages, which are outlined below.

1. Define a method for automating a document preservation process, specifically appending metadata to a large set of files.
2. Define an automated method for multilingual (i.e. French, Vietnamese, and Chinese-Traditional) optical character recognition (OCR) and translation.
3. Generate a reference database of keywords from *L'Art à Hué* to use in assigning labels and subject metadata to the Vietnamese stele images and the L'Art à Hué scans, respectively.

3.2.1 Appending Metadata to Files for Document Preservation

After researching document preservation methods used by the Library of Congress and the United Nations, we opted for the Dublin Core metadata standards and TIFF file format (Library of Congress, n.d.-a; United Nations, n.d.). A Tag Image File Format (TIFF) is a computer file used to store pixel-based raster graphics and image information (Adobe, n.d.-f). TIFF files implement a lossless compression, meaning they maintain the original image quality but take up more storage space on a device (Adobe, n.d.-f). Because the scans of *L'Art à Hué* are purposed for document preservation and reconstruction, the image quality is important and cannot be sacrificed. We explored methods for embedding metadata via the Dublin Core standard into the individual files. As discussed in Section 2.2.1, Dublin Core is an internationally agreed-upon standard for metadata information that should be included in a stored file.

After comparing multiple software platforms capable of processing the images based on our requirements, we identified and implemented Adobe Bridge for its accessibility, adaptability, and batch automation features (Adobe, n.d.-c). We found that we would be able to implement custom scripts to automate the processing procedure and add the metadata to the full set of images in only a few hours. Since our goal was to create a generalized set of steps that could be used by historians in the future, it was a priority to establish a short and simple process.

We built the rest of the procedure around the utilization of Adobe Bridge to append the metadata to the TIFF files. The files were originally provided to us as PNGs. Adobe Bridge was unable to natively convert PNG files to TIFFs, so we used the built-in connection to Adobe Photoshop to save the full batch as TIFF files. Much of the metadata was the same across the images, so we were able to code a custom template in Adobe's Extensible Metadata Platform (XMP), which is a file labeling technology that allows users to embed metadata into the files they are working with (Adobe, n.d.-e). Creating the template allowed us to quickly append the consistent metadata to all images in a batch process. We included all of the Dublin Core data fields in the template and left the image-specific fields blank to be filled in later. Using an open-source toolkit called Metadata Deluxe, we were able to export the metadata for all the files in the batch to a *.txt file that we could view in Excel (Metadata Deluxe, n.d.). In this format, we could edit each field individually and import the metadata back into Adobe Bridge to fill in the missing fields.

All of the metadata fields, with the exception of the subject field, were completed without additional processing. The captions were included in the description field in both the French and English. The translation was executed using Google Translate (Google LLC, n.d.). To finalize the subject field, we first needed to finish the reference framework, which we later discuss in Section 3.2.3, to establish a list of possible keywords to use across the images. Once

that list was complete, we entered the keywords that corresponded with the subject(s) of each page in the subject metadata field in the *.txt file. This included all of the referenced entities on the text pages and all of the visual entities on the art pages. We then used the Metadata Deluxe toolkit in Adobe Bridge to import the complete list of metadata and append it to the images.

The only remaining task was creating a consistent file naming scheme. We decided to use the identifier field, which specified each image’s origin chapter and page number as well as a one-word description of its contents. We encountered some discontinuities between images that were not included in the PDF scan of the book and vice versa, so we created an error-code that indicated there was an issue between the two versions by numbering the page as zero (i.e. “p0”). We then used a custom script, which was generated by Google Gemini and manually tested and debugged, that renamed all of the files to their associated identifier automatically.

3.2.2 Performing Multilingual OCR and Transcribing Textual Documents

To analyze the content of our reference text, multiple processing steps were performed on the digitized version of *L’Art à Hué* initially provided by our sponsor. First, the TIFF files with custom metadata (see Section 3.2.1) were processed using ScanTailor and exported again as TIFF files. The processing included fixing the DPI to ensure accurate digital recreations, deskewing, and binarization. ScanTailor is a standard tool used for batch image processing of this type and was a recommendation by our sponsor (ScanTailor, n.d.).

The preprocessed TIFF files then had their contents extracted by uploading them to Google Gemini, which performed the Optical Character Recognition (OCR) as directed by a standardized prompt (see Appendix C for LLM prompt details). The format of choice for the OCR output was as plain text, which, unlike a PDF, is easily manipulated to correct for formatting inconsistencies. Only the “text pages,” or pages that consisted mostly of text with minimal images, were subjects of OCR. The captioned “artwork pages,” shown in Figure A, had only redundant information, and the associated caption content was already manually added to their metadata (see Section 2.2.1).

Restricting the batch size for the OCR process to a smaller subset of the files had the benefit of significantly reducing both the human workload and the LLM usage. A second prompt was then used to produce an exact English translation of the French text, which was manually analyzed to find the entities of interest for the authority file (see Appendix C for LLM prompt details). The Vietnamese and Chinese characters were left unchanged.

Google Gemini was chosen for its ability to perform the OCR and translations, as well as its flexible token limit that allowed us to process the pages individually. However, before deciding on this tool, multiple alternative approaches were explored. Non-LLM solutions included NAPS2, Adobe Acrobat, OCRmyPDF, and ABBYY FineReader, each of which had unique drawbacks that produced unsatisfactory results (Olden-Cooligan, n.d.; Adobe, n.d.-a; Barlow, n.d.; ABBYY, n.d.). Other LLMs were tested as well, such as Qwen, DeepSeek, and Microsoft Copilot, but each of these also generated inferior results in the OCR and translations compared to Google Gemini. Even when presented with faded or unclear characters, Google Gemini was able to identify the unclear text based on surrounding context and output an exact transcription or translation.

3.2.3 Creating an Authority File Database

The Object-Oriented Database

With the reference document processed and the text extracted, the next step towards a fully automated classification pipeline was constructing an authority file. Since an authority file is a type of database (see Section 2.2.3), we began by interviewing computer science professors at Worcester Polytechnic Institute that specialize in databases. We interviewed two professors, who each separately confirmed that an object-oriented database model would best represent our data. In a traditional entity-relationship model, as shown below in Figure B, entities are grouped into defined categories with pre-defined attributes and relationships (Talbut, 2011).



Figure 6: A screenshot of an “artwork page,” HUE-ch5-p238-planche156, from *L’Art à Hué* (Cadière, 1919). The artwork pages contain one or more artistic works that are described by the caption, which provides a unique number that identifies the piece within the book and a short description of what the piece is.

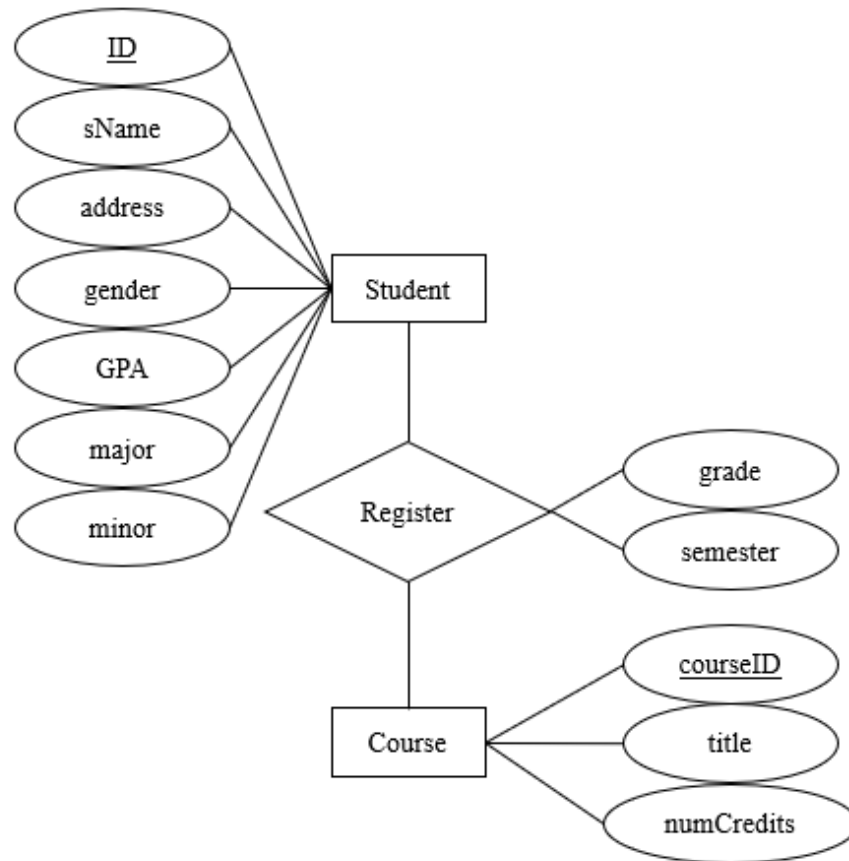


Figure 7: An example of an entity-relationship database model. Rectangles are entities, ellipses are their attributes, and diamonds are relationships between them. In this example, each entry into this database would be either a student or a course (i.e. as rows in a table). Each would be associated with a list of the attached attributes (i.e. as columns in a table) and related to another entity.

The entity-relationship model is useful in cases where every instance of an entity will have the same set of attributes and relationships. This model would not have worked for our data because many of the entities were organized in a hierarchy and had inconsistent relationships with other entities across the full dataset.

In an object-oriented database, each entity, or node, is its own unique object with a unique set of attributes and relationships (Ferguson and Hebels, 2003). This creates a more graphical representation, as shown in Figure C, where any node can be connected to any other node without restrictions.

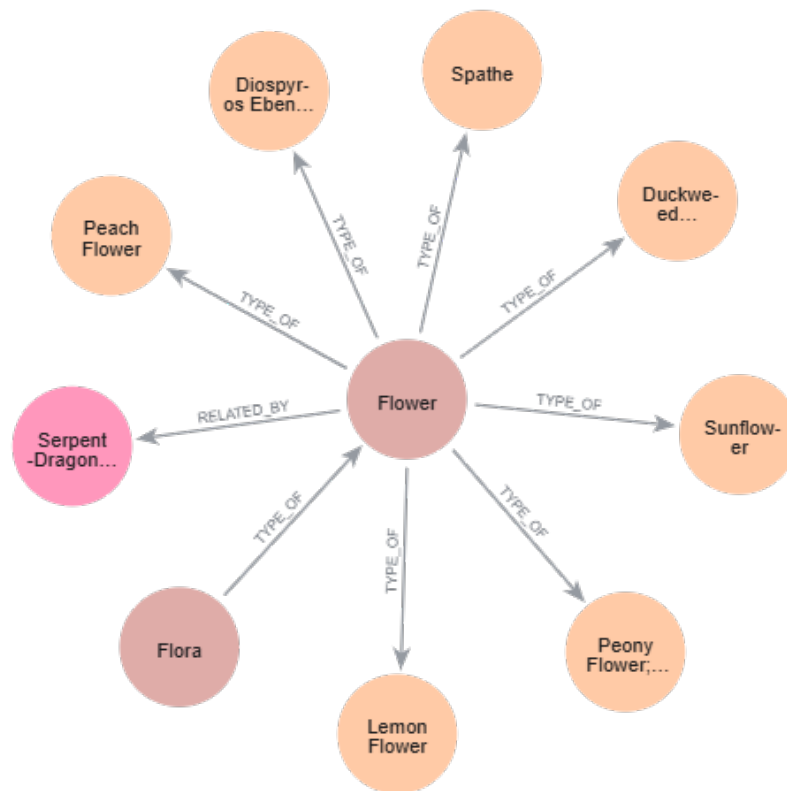


Figure 8: An example of an object-oriented database model. Each circle represents a unique entity, or node, and the arrows are relationships between them. This is different from the entity-relationship model because each node has a unique set of attributes and relationships.

The professors recommended a software called Neo4j, which allows for seamless graphical database organization and management (Neo4j, Inc., 2026).

Creating the Nodes

Before entering the data into Neo4j, we needed a list of entities and relationships. We read the English text transcription of L'Art à Hué that was generated via the process outlined in Section 3.2.2 and compiled a list of art objects that could potentially be depicted on the Vietnamese steles. We compiled this list in Google Sheets and specified the English, French, Vietnamese, and Chinese terms as well as the high-level relationships between the objects.

For objects that had alternate names, we listed them all in the appropriate cell and separated them with semicolons. We verified the differentiation between the Vietnamese and Chinese terms with Google Translate and added broad categories to each object based on the chapter titles (Google LLC, n.d.). With all of the entities, relationships, and categories defined in the table, we could move on to constructing the database in Neo4j.

Constructing the Database

To keep the database organized and at a minimum complexity, we outlined a specific set of possible categories and relationships that we applied consistently across the database. The categories represented different hierarchy levels of the database, such as Animal, Supernatural Animal, and Dragon, which were all included as node categories. Each node also included five property keys, or information fields, which were the English, French, Vietnamese, and Chinese translations of each node and a default key, which was the English title of the entity. For the full list of categories, relationships, and property keys, refer to Appendix D.

Database platforms use query languages to construct, reference, and modify their entries (Reisner, 1988). Neo4j has its own declarative graph query language, called Cypher, that provides queries for property graphs (Neo4j, Inc., n.d.-a). Using Cypher, we implemented a series of MERGE statements that created nodes and relationships based on our entries into the Google Sheet. We wrote the queries in a TXT file before pasting them into Neo4j and making modifications as necessary.

Once the database was complete, we had a reference framework with terms in English, French, Vietnamese, and Chinese that fully encompassed the nineteenth-century Vietnamese entities and relationships defined in *L'Art à Hué*. The goal was to use this framework as the official list of possible terms for the subject metadata field in our scans and for classifications with the artificial intelligence model.

3.3 Project Objective 2: Creating an AI Pipeline for Image Recognition and Classification

The second project objective was to create a pipeline integrating artificial intelligence for image recognition and classification. We accomplished this by implementing and fine-tuning foundation models.

3.3.1 Foundation Models

Foundation models are artificial intelligence models that have been pre-trained on large datasets with flexibility in mind, allowing them to be applied to a wider range of tasks than

a typical machine learning model designed and trained for a specific implementation. Our team employed a series of various foundation models as a means of detecting, classifying, and segmenting the objects or motifs depicted on the Vietnamese steles. The first of the foundation models in the pipeline was Qwen3-VL-8B, a Visual Language Model (VLM) developed by Alibaba Cloud with the training to handle a broad range of tasks (QwenTeam, n.d.). Qwen is a superior choice compared to other large mainstream models as its training data emphasizes image analysis (QwenTeam, n.d.). VLMs such as Qwen3-VL are highly capable of perceiving visual information and performing reasoning and analysis with the provided visual content (R. D. C. Stryker, 2025). Comparatively, models like OpenAI's ChatGPT are trained on large amounts of text and code with less emphasis on image inputs, defining them as Large Language Models (LLM) (C. Stryker, 2021) . While typical AI models or popular LLMs, such as ChatGPT 5.0, may be able to adequately handle the tasks we desire when given the correct prompting, using models like Qwen made this process easier, more accurate, and facilitates simpler replicability of the foundation model based pipeline.

In addition to the usage of Qwen, our team experimented with more specialized foundation models, including GroundingDINO, Florence2, and SAM2. GroundingDINO and Florence2 are models designed specifically for object detection within images (Pacheco and Rogge, n.d.). This capability highly compliments SAM2, which is an image segmentation model (Meta AI, n.d.). Pipelines, such as GroundedSAM, which is the combination of GroundingDINO and SAM2 for object detection and segmentation, explore combinations of these models to perform more specialized tasks which coincidentally accomplish the majority of the intended AI-based object detection pipeline's purpose (Alibaba Cloud, n.d.).

Our team looked into three pipelines with the goal of determining the ideal combination of foundation models to implement in pursuit of accurate and reliable motif detection on the Vietnamese steles. These pipelines include GroundedSAM, Florence2 combined with SAM2, and Qwen3-VL-8B alongside SAM2. All three pipelines have been implemented and evaluated in the same manner, involving the same input image of a stele and adjusting the parameters of the models to determine the capabilities of each pipeline. Due to the difference between the models, they have been tested using the same object of interest, for instance dragon motifs, but their respective detection parameters were specifically tailored to each model in order to give the best possible results for comparison.

The creation of these pipelines, specifically the implementation of Qwen3-VL and SAM2, involved the adaptation of pre-existing code written by a group of creators labeled IDEA-Research. This group assembled the GroundedSAM2 pipeline, a publicly available object detection and segmentation code involving the use of GroundingDINO and SAM2, as previously detailed (Tianhe and Shen, n.d.). Due to the identical end goal and simple substitution

of duties, the original GroundedSAM2 pipeline python code was adapted to utilize a locally ran instance of Qwen3-VL-8B rather than the intended GroundingDINO. This alteration in pipeline code was performed by hand with assistance from Anthropic’s Claude Sonet 4.6 model to ensure seamless implementation.

Model Fine-Tuning and Prompt Engineering

When initially implementing the foundation models for object detection and image segmentation, the outputted results were substandard compared to what our team desired as a final deliverable. These inadequacies were in large a result of the task at hand, correctly detecting and classifying Vietnamese motifs from scans of stele rubbings. Due to the esoteric nature of the stele’s images and the generalized, or real-world centric, dataset that the object detection models have been trained on, there exists a disconnect between the models’ capabilities and the tasks required (Picsellia Team, 2023) . To put it in perspective, the object detection models excel at picking out a naturally existing object, such as a car or dog, from a provided image or photograph due to their prevalence in the model’s training data. When the same model is asked to locate a scanned rubbing of a damaged seventeenth century stone carving of a stylistically depicted dragon, it struggles and either misses the desired object or hallucinates an output where none exists.

To combat the shortcomings of the models, the specific prompts provided as well as the detection parameters were fine-tuned to optimize the models’ processes and resulting outputs. For the detection models, GroundingDINO and Florence2, parameter adjustment was the main method of optimization. This is due to these models inability to take extensive prompts as inputs, meaning their task specifications were limited to simple object names and descriptions. The inability to accept complex text prompts, as well as their disadvantages in reasoning, is what led to the adoption of Qwen3-VL for object detection. Qwen3 offers a far greater capacity for optimization, allowing for complex text prompting, complex image descriptions, equal if not superior model parameter tweaking, and additional semantic reasoning (Qwen, n.d.).

The process of fine-tuning the models was performed using a standardized input with the goal of detecting and segmenting the same objects from within the image. For testing purposes we selected a pair of steles from Vietnamica, an image database that contains the 20,980 rubbings of steles produced in Vietnam by the École Française d’Extrême-Orient (Vietnamica Gallery, n.d.). These steles were numbers 10925 and 11170, displayed as Figures 9 and 10 respectively. These steles were chosen specifically as they cleanly display a variety of desirable motifs with a range of image quality, location, and complexity. Using these control images, the models were then tasked with identifying the depicted objects over a

sweep of box and text threshold values. These threshold values within the code, box and text, control the minimum identification confidence required by the machine to propose a bounding box and determination of the detected object respectively. By performing the same detection over a range of threshold values, from 0.05 to 0.35, we were able to determine the ideal values for accurate detection. At values of 0.15 for both the box and text thresholds, the model detected the most objects from within the image of the stele without hallucinating and creating false positives. At low values, 0.10 and below, the models began sporadically selecting objects from on the stele and labeling them with random object names. The inverse happened at thresholds of roughly 0.2 and above, with the models missing clearly depicted objects and ignoring much of the stele's content.



Figure 9: Stele #10925 (Vietnamica Gallery, n.d.). This rubbing is of a well preserved stele, chosen for its dragon and ornamental depictions on both the crest and bottom as well as the quality of the rubbing and object depictions. and relationships. Due to the comparatively simplistic depictions and low variety of pictured objects this stele was initially chosen to focus on the detection of a single object, dragons.



Figure 10: Stele #11170 (Vietnamica Gallery, n.d.). This rubbing of stele 11170 was chosen due to its increased visual complexity and variation in depicted characters compared to the previous stele, 10925. Still maintaining a higher quality of image preservation ideal for image detection, the objects contained on the stele provide variety to better test the flexibility and capability of the image detection pipelines.

Prompt engineering, or optimizing the textual input for the model, is performed by specifying what input and direction is given to the AI model (Amazon Web Services, n.d.-b). By using Qwen’s vision language model, we were able to greatly increase the depth of the text prompts beyond the capabilities of other tested object detection models. The reasoning capabilities of Qwen allot for more specific object descriptions and instructions, which assist in the detection of unique objects, such as the motifs depicted on the Vietnamese steles. The GroundedSAM and Florence2 + SAM2 pipelines only allow for a minor amount of prompt engineering, limited to the description of the object that is being requested for detection. For these simpler models the prompts are limited, for instance, to describing a baseline request of “dragon” to something more specific, such as “left facing dragon motif.” However, even with such a simple description the quality of the output remained relatively poor. In comparison, the input for dragons in the Qwen prompt contained multiple lines of text describing the

object’s appearance, its typical location, the key components of a dragon motif, etc. In addition to this exponential increase in prompt specificity, the model is given an overarching prompt detailing its role in the pipeline with extensive textual input explaining the context of the work with intricate instructions for what exactly the model is being instructed to accomplish, something not possible with the GroundingDINO and Florence models.

Quantitative Analysis

To quantitatively evaluate the final model’s performance, our team used the Mean Average Precision (mAP) score. The mAP score is a commonly used metric that measures the accuracy of object detection models, accounting for all Confidence Thresholds, by averaging Average Precision (AP) scores across all classification categories or classes (Anwar, 2022). mAP is found using the following equation:

$$mAP = \frac{1}{k} \sum_i^k AP_i \quad (1)$$

In the above equation, k is the total number of object classes and each AP_i is an Average Precision associated with one of the k classes.

To get each AP score, multiple other statistics must be defined. The first of these is the Intersection over Union (IoU), which is the ratio between shared area and total area between ground truth and prediction bounding boxes:

$$IoU = \frac{A_{\cap}}{A_{\cup}} \quad (2)$$

Here, $A_{\cap}$ is the area of intersection between the ground truth and prediction bounding boxes and $A_{\cup}$ is the area of union, or total area, between the two. IoU lies in the interval $[0,1]$, with a score of 1 indicating perfect overlap and 0 indicating no overlap. The IoU value for each prediction (or lack thereof) is used to determine true positive, false positive, true negative, and false negative counts by comparing the IoU to a set of predetermined confidence thresholds (Anwar, 2022). The figure below outlines the definitions of true positive, false positive, true negative, and false negative. See Figure 11 for more details.

With true/false positive and negative counts established, two more values, precision and recall, can be found for each class and confidence threshold:

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

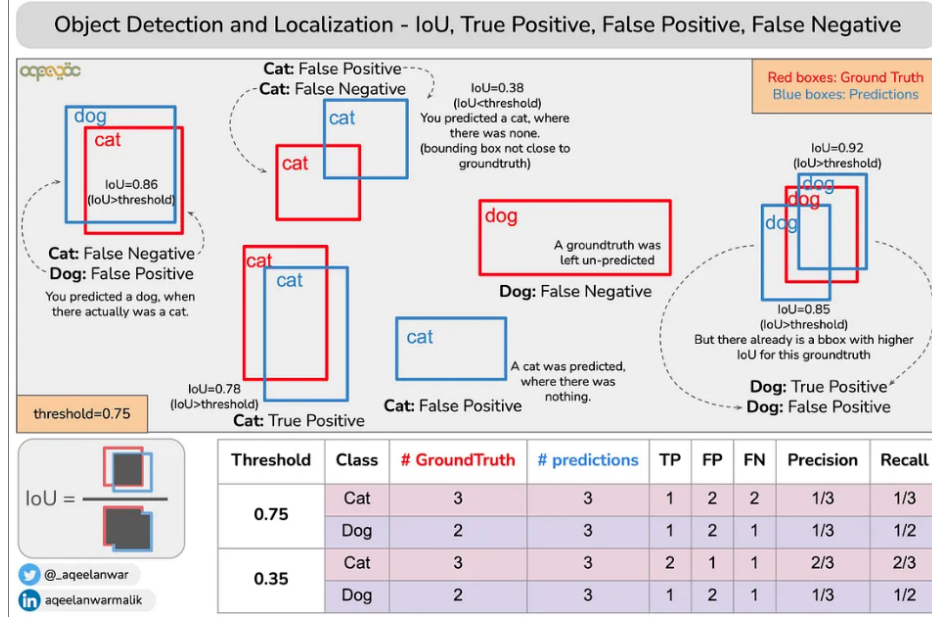


Figure 11: An infographic detailing the process of determining true/false positives and negatives. It uses the example of classifying cats and dogs, and shows how to compare predictions against a human-made ground truth to determine how each prediction should be classified. It also includes a small diagram that visually explains Intersection over Union (Anwar, 2022).

TP is the true positive count, FP is the false positive count, and FN is the false negative count (Anwar, 2022). Because the denominator of each fraction is always greater than or equal to the numerator, precision and recall are both proportions, lying in the interval $[0,1]$.

With precision and recall values found, a precision-recall (P-R) curve can be generated by plotting precision against recall. P-R curves show how these values change with different confidence thresholds for a given class (Anwar, 2022).

Using each P-R curve, the average precision for its associated class can be found by taking the integral of the P-R curve on the unit interval:

$$AP = \int_0^1 p(r)dr \quad (5)$$

$p(r)$ in Equation 5 is the function for the P-R curve. However, precision and recall are calculated for a discrete set of confidence thresholds, so approximation methods such as Riemann sums are often preferred over continuous integration. AP lies in the interval $[0,1]$, with higher scores indicating a better average precision (Anwar, 2022).

Finally, mAP can be found by taking the arithmetic mean of AP scores across all classes (eq. 1). Because mAP averages AP scores, which already contain information on different confidence threshold performance, across all classes, mAP provides a well-rounded statistic to evaluate how accurately classifications were made (Anwar, 2022).

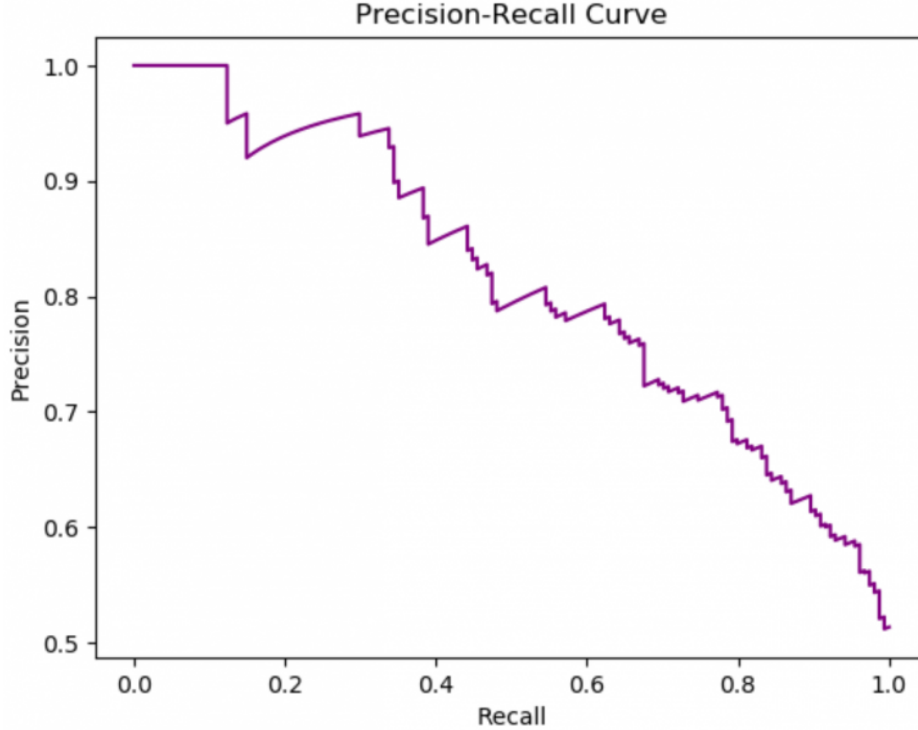


Figure 12: An example of a Precision-Recall Curve for a prediction model. Recall is plotted along the x-axis and Precision along the y-axis. Both lie in the interval $[0,1]$ (Bobbitt, 2021).

4 Findings and Deliverables

In this chapter, we detail the findings from our expert interviews, as well as our proposed pipelines for data organization and management and automatic image classification.

The documentation, code, output, and other resources produced by our methods can be found on the GitHub repository at this link: <https://github.com/wpilyone26archeology/automation-of-methodologies-for-archival-processes.git>. The file structure can be found in Appendix E.

4.1 Expert Interview Findings

All of the interviewees provided similar commentary on the functional aspects of an AI model for augmenting the image classification of steles. Our line of questioning was tailored towards the nature of steles and AI technology, this generalizable approach allowed the information from the interviews to inform the scope of our project.

The computer science professors emphasized that the computational side of stele analysis is inherently difficult due to the nature of the artwork on steles. They highlighted that it would be very difficult to augment a classification process with AI because there is so

much creative style variability within the artwork. The uniqueness of not only the artwork on steles but the reference text used to analyze them makes the characters functionally non - standardizable. An AI model would need a much larger training dataset to recognize individual motifs than exists within this corpora. This truth pivoted the focus of our study away from building a model, and towards finding a way to correctly preprocess and label the dataset. The emphasis the computer science experts put on this step was what defined our project scope. Creating an AI model to augment archaeological processes would be very complex, niche, technically difficult, and time consuming. Identifying a generalizable process of adapting datasets to be compatible with future AI use became the object of our study.

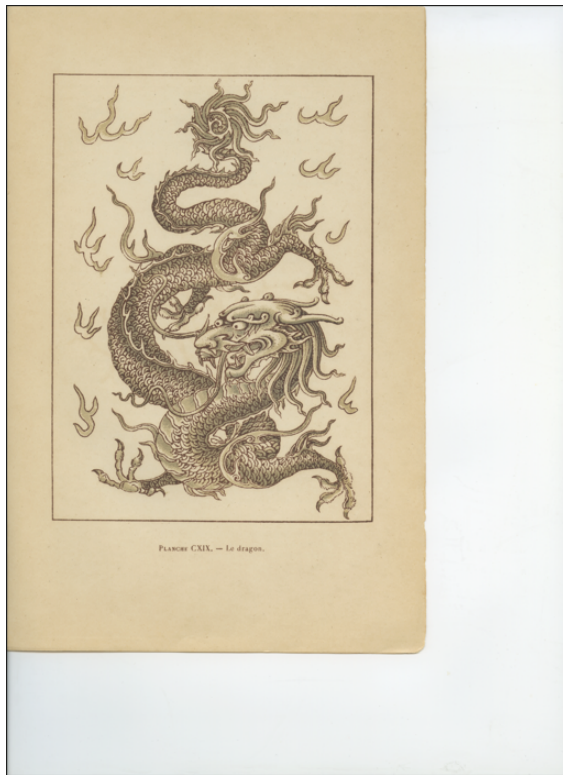
These technical interpretations given by computer scientists were supported by the archaeologists and historians we interviewed as well. In trying to understand the patterns and predictability of the steles, we found that the meaning behind them is far more intricate than we are capable of understanding without an extensive background in Vietnamese culture, religion, and history. Slight visual variations within the individual motifs represent different metaphysical definitions all together. The emphasis these experts put on the complexity of interpreting the steles further helped to define the scope of our project away from classification of steles and towards proper document preservation and database establishment.

4.2 A General Pipeline for Archaeological Data Organization and Management

Using the tools discussed in the methodology section, our team generated the “Method for Appending Metadata to Files for Document Preservation,” outlined in Appendix X, to convert the 212 images of *L’Art à Hué* to TIFF files and append their corresponding metadata. The full table of metadata can be found on the repository, but a summarized version of the Dublin Core data fields is included in Appendix G.

The output files were then processed using ScanTailor to produce normalized images, following the procedure outlined in the first section of the “Method for Performing Multilingual OCR and Transcribing Textual Documents,” included in Appendix F. Figure 13a and Figure 13b below show the results of the conversion and normalization, respectively.

Our team then produced the English and French transcriptions of the text using Google Gemini and the procedure in the second section of the “Method for Performing Multilingual OCR and Transcribing Textual Documents,” included in Appendix I. After reading these transcriptions, we produced a database following the “Method for Creating an Authority File Database,” outlined in Appendix J, which contained all of the entities and the high-level relationships depicted in the text. The full graphical representation is shown below in Figure



(a) A screenshot of the processed TIFF file, HUE-ch5-p196-planche119, that contains metadata with associated values from Table 1 (Cadière, 1919).



(b) A screenshot of the TIFF file, HUE-ch5-p196-planche119, after being processed by ScanTailor (Cadière, 1919).

Figure 13: Results of the two intermediate stages of the data organization and management pipeline.

14.

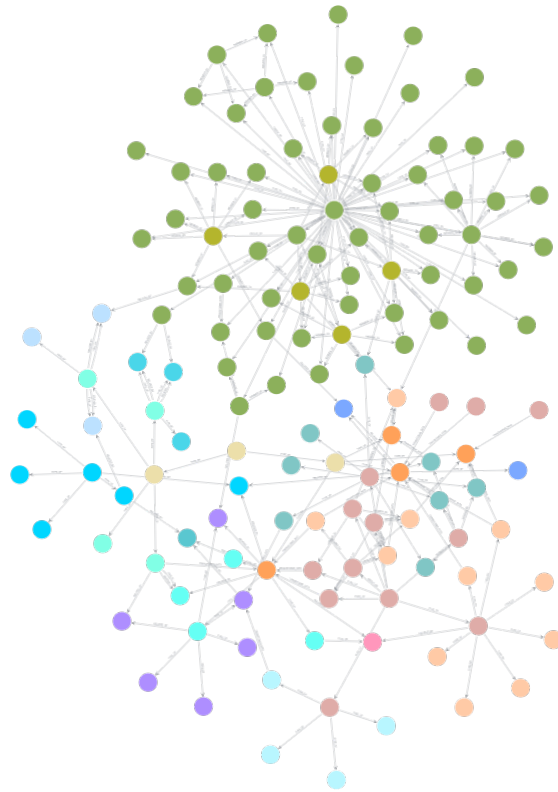


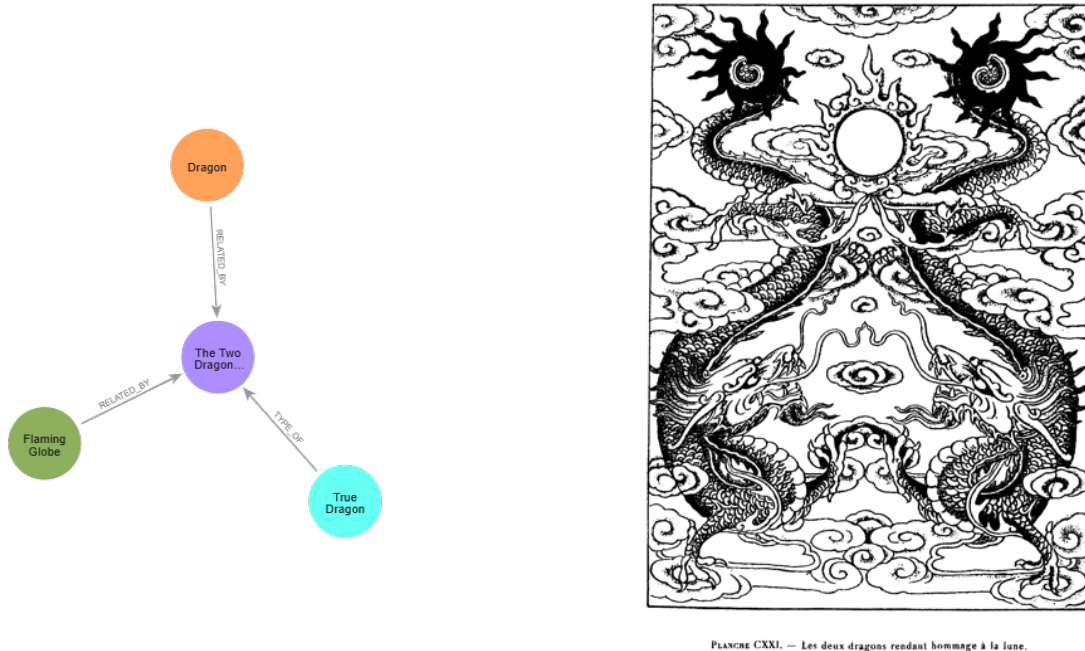
Figure 14: An image of the graphical representation of the authority file created by reading the English transcriptions of *L'Art à Hué*.

The code used to reconstruct this database in Neo4j can be found in the repository, along with the table of relationships. Although our team outlined only the high-level relationships between entities, the database became very complicated very quickly. The selection of which relationships to include was completely subjective and was based on two key factors. The first factor was whether a relationship was hierarchical. Many entities were subsets of others, like different types of flowers. After interviewing two specialists in Vietnamese history, we determined that the most useful implementation of a classification model would allow historians to filter through and search for these types of relationships in order to form conclusions and find patterns within the dataset. The second key factor was whether or not a relationship created a new entity. The graph in Figure 15a shows this type of relationship.

The purple node is a piece called “The Two Dragons Paying Homage to the Moon,” which includes a depiction of two dragons and a flaming globe. The associated artwork is shown in Figure 15b.

This piece would not exist without both the dragon and the flaming globe entities, which was why it was important to denote these types of relationships in the database.

After creating the full database, we were able to complete the subject field in the metadata and append the final set to the TIFF files. The appended metadata provides information about the contents of each image and information about their origin, which enables both streamlined database organization and classification, as well as reconstruction of the original document in the future.



(a) An image of an example of the graphical representation of a relationship that created a new entity in the database. This example shows "The Dragon Paying Homage to the Moon," which is created when the Flaming Globe and two True Dragon entities are depicted together (Cadière, 1919).

(b) A screenshot of the TIFF file, HUE-ch5-p198-planche121, which depicts "The Two Dragons Paying Homage to the Moon" (Cadière, 1919).

Figure 15: Example of instances when combinations of entities create new ones.

To view the full database on Neo4j, refer to the access codes and instructions outlined in the Interactive Example section of Appendix J.

4.3 An AI Pipeline for Image Recognition and Classification

Using the tools and processes detailed within the methodology section, we have established a theoretical pipeline that can be used to detect and segment motifs from within scans of the steles. While imperfect due to time, cost, and technological constraints, the finalized process,

recommended technological implementation, and next steps for improvement of said process are expected to deliver an idealistic AI based image recognition and classification pipeline.

One of the largest determinations made through our work in creating the AI pipeline is that the use of personal computers for this work is inadequate and proves to be a limitation to both the quality and extent of the work at hand. The computational demands of running locally installed AI models exceeds that of the average personal computer's capabilities, most notably if one wishes to employ the ideal software for the task at hand. Either utilizing a far more capable system or outsourcing the computational demand through paid API usage will far greatly improve the quality of outputs compared to that capable when ran locally on a personal computer. Displayed below in Figure F, is the output of the final Qwen3-VL-8B & SAM2 pipeline. It is apparent that the object detection and segmentation of said objects leaves much to be desired and displays significant room for improvement.

4.3.1 Quantitative Analysis

As discussed in the methodology section, we quantitatively analyzed the model's performance using the Mean Average Precision (mAP). Across all trials, the model could not produce predictions that yielded a mAP higher than 0.0000, which is the lower bound for this metric and indicated no accurate classifications. This was most likely due to the size of the prediction bounding boxes that it generated; none of them fully encompassed whole motifs, so the Intersection over Union was always much less than one. This resulted in no predictions being labeled as true positives by the program, so both precision and recall and, consequently, mAP were zero.

5 Conclusions and Recommendations

Our team addressed the complexity of the archaeological classification process by creating a series of three pipelines to serve as the basis for an artificial intelligence (AI)-based automated procedure. The first pipeline was the "Method for Appending Metadata to Files for Document Preservation" (Appendix F). This process involved appending metadata in the fifteen-element Dublin Core standard to archaeological files stored in a lossless compression TIFF format. Doing so enabled both automatic classification and simplified data organization based on the metadata, as well as future document reconstruction using the preserved quality of the TIFF files. The "Method for Performing Multilingual OCR and Transcribing Textual Documents" (Appendix I) built upon this procedure by creating machine-readable transcripts from the TIFF files that were later translated into multiple languages. This multilingual feature was a major component of the final pipeline, the "Method for Creating an Authority File Database"

(Appendix J). This method resulted in a graphical database that defined entities and the relationships between them based on the textual transcriptions. The translations allowed for the associations of multilingual keywords for each entity, enabling an expanded and inclusive vocabulary within the database. Each process was tested using PNG scans of the pages from the nineteenth-century Vietnamese artbook, *L'Art à Hué*, and were all successful (Paris Musées, n.d.-b).

The three processes created the beginning of a normalized standard which our team used to explore an AI pipeline for the classification of archaeological artifacts. Using scans of seventeenth-century Vietnamese stele rubbings, our team implemented a pipeline using Alibaba Cloud's Qwen3-VL-8B and Meta's SAM2 (Qwen, n.d.; Meta AI, n.d.). Although the results of the tested classifications were unsatisfactory due to time, cost, and technological constraints, the integration of AI tools in archaeological classification should continue to be explored to assist in this complex process.

Based on our findings, our team recommends three expansions to these pipelines. First, because there are currently no AI tools capable of performing Optical Character Recognition (OCR) in French, Vietnamese, and Chinese without the implementation of a Large Language Model (LLM), the OCR was completed using Google Gemini, Google's LLM. Based on the ethical considerations discussed in Section 2.4, we determined that fine-tuning an existing OCR model or creating one from scratch capable of all three languages would be beneficial, as it could run locally and reduce the environmental impacts generated by prompting a large LLM like Google Gemini.

The second recommended expansion is designing an automated pipeline for the creation of the authority file. The current methodology requires knowledge of Cypher, Neo4j's query language, to complete the database (Neo4j, Inc., n.d.-a). Streamlining the process to an automated procedure would allow experts to implement the method without a coding knowledge requirement and would accelerate the completion of the database.

Our final recommendation is to train an AI model on the dataset of Vietnamese stele rubbings. The SAM2 model was trained on images of real objects, such as locations, objects, and scenes (Meta AI, n.d.). The motifs depicted in the margins of the steles therefore presented a challenge, as the model had never encountered these types of objects before. Exposing the model to a variety of Vietnamese symbols and motifs would greatly improve its capability in identifying and classifying these types of objects. The artwork in *L'Art à Hué* could potentially serve as a ground truth for the classifications. Alternatively, implementing few-shot learning with one to five reference images of each motif from the set of Vietnamese steles could serve as a ground truth and abbreviated training procedure. Few-shot learning (FSL) is a framework in which an AI model learns to make accurate predictions by training

on a very small number of reference images (IBM, 2024). Either method could prove to be successful, so exploring the accuracy of both would be required.

The resulting methodologies and recommendations produced by this project serve as the groundwork for future implementations of automation in the field of archaeology. Automation using artificial intelligence streamlines complex procedures and tasks while serving as a tool for experts that assists in their workflow. Building from this groundwork will provide assets for archaeologists to expand their work further beyond data management and organization and towards uncovering the history of past civilizations.

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Appendix A: Sponsor Description

The sponsor for our project is Professor Marc Bui, a computer scientist at Université Paris 8 Vincennes-Saint-Denis. Professor Bui is also the Director of Cumulating Studies at École Pratique des Hautes Études (“Marc Bui - Archaeology & Philology of the East and West - CNRS PSL”, n.d. He is an expert in his field, with extensive experience showcased in his contributions to over 20 publications, including conference papers and journal articles (“Marc Bui - Archaeology & Philology of the East and West - CNRS PSL”, n.d.

Professor Bui’s experience with machine learning and artificial intelligence, data augmentation and automatic transcription, and prior experience and expertise in ancient steles are essentially the backbone of our project. Professor Bui has been working with Vietnamese steles to develop a model that can autonomously decipher the textual content of a stele. The work of Bui and his associates is substantial, however it still leaves room for progress with the goal of improving precision and expanding the scope of their work in the future (Scius-Bertrand et al., 2023. His research will prove as a solid foundation for what our project team aims to accomplish, incorporating artificial intelligence to classify components of these archeological figures.

Appendix B: Interview Questions

English Version

General Questions

1. How do you feel about the term “digital humanities” and the use of artificial intelligence (AI) for automation within your field of study?
2. How important is understanding the cultural and historical significance of a project such as ours, and how important is it to convey this significance in the final product?

Questions About Buddhist Steles

Representation, Historical Context, and Interpretation

1. How thoroughly do steles represent the lives of the individuals for whom they were created?
2. Are steles accurate representations of individuals’ lives, or do they tend to exaggerate certain aspects of their identities, accomplishments, or status?
3. Is the information contained within steles extensive enough to provide a meaningful understanding of an individual’s life and accomplishments, or do they primarily serve as brief summaries?
4. What is the importance of studying economic history through steles today?
5. How much does the historical information depicted within steles differ from “official” records created by governments or officials during the same period?
6. How important is the date of a stele’s creation when interpreting its meaning? Could interpretations change significantly depending on the historical period?
7. How much variation exists between steles from different regions? Is it possible to generalize the meaning of symbols when those symbols may have had different interpretations across regions or periods?

Artistic Style, Symbolism, and Classification

1. Are there distinct regional artistic styles within depictions of the same animal or motif that should be treated as separate classes?

2. What Buddhist concepts are most important to understand when analyzing steles (e.g., karma, impermanence, cosmology)?
3. How does Buddhist cosmology influence the spatial organization and orientation of elements within steles?
4. Were there periods of stylistic de-standardization (such as the 17th or 18th century) when established artistic conventions were intentionally altered?
5. How can we ensure that an AI model does not classify creative stylistic variations or artistic outliers as errors or separate categories?
6. In Buddhist steles, does every animal or depicted object carry semantic meaning, or are some elements primarily decorative?
7. When a dragon and phoenix appear together, does their combination alter the meaning of each individual symbol compared with when they appear separately? Should this relationship be treated as a separate classification?
8. How should hybrid creatures (for example, a dragon transforming into a cloud) be classified? Should they be treated as separate classes or as attributes connected to a primary category?
9. When identifying ornamental elements, is recording their position on the stele a necessary attribute?
10. Many ornaments appear to communicate complex historical narratives, including sacred status, historical legitimacy, social rank, and artistic trends. Can interpretations of these narratives be effectively outsourced to AI, or should AI primarily assist with organizing datasets while interpretation remains the responsibility of human experts?

Questions About AI and Historical Interpretation

1. When provided with sufficient contextual information, can AI be used to interpret historical data accurately?
2. Can AI independently create reliable interpretations of historical materials, or is human expertise always necessary?
3. If AI were theoretically capable of interpreting historical documents and artifacts as accurately as, or more accurately than, human experts, would AI become the preferred method of analysis? Or should human scholars retain final authority over interpretation?

4. Since AI models can contain inherent biases, what aspects of Buddhist stela history, symbolism, or cultural representation are most likely to become underrepresented?

Questions About Our Project and Data Modeling

Representation and Relationships

1. How can we ensure that less-visible, marginalized, or underrepresented historical perspectives are included in our dataset?
2. How are interpretations of steles used today? Are they connected to modern issues such as cultural identity, historical claims, or land disputes?
3. How do researchers determine relationships between entities represented in archaeological and historical datasets?
4. When segmenting stela images, should AI models classify isolated motifs (e.g., a single dragon claw) or holistic compositions (e.g., a dragon chasing a pearl)? Which unit of analysis remains more stable across different centuries?
5. What common motifs, relationships, or symbolic connections found in steles should we be aware of, and how can we successfully identify them?

Question About Authority Files and AI Training for Cultural Heritage Objects

Metadata and Structure

1. What are the most important properties that must be included in an authority file for archaeological objects to make it useful for AI training?
2. How should variability in terminology (e.g., regional names, synonyms, historical terms versus modern terms) be managed within a controlled authority file?
3. What is the appropriate level of specificity for classifying object types for effective machine learning (e.g., “container” versus “amphora” versus “wine amphora”)?
4. How should uncertain or partial identifications (e.g., “possible brooch” or “fragment of unknown object”) be represented without confusing an AI model?
5. What is the best practice for representing hierarchical relationships between object categories (e.g., ceramic → pottery → bowl) in a machine-readable format?

6. How should temporal and cultural variations in object classification be managed when the same object type may have different names across periods or regions?
7. What metadata should accompany each authority term (e.g., definitions, images, bibliographic references, confidence scores) to improve AI training quality?
8. What are the most common mistakes or challenges you have observed when creating authority files for cultural heritage AI projects?
9. If you could design an ideal authority file structure specifically for AI training today, what would you change compared with existing standards such as the Getty Art & Architecture Thesaurus (AAT) or Dublin Core?

Version Française

Questions Générales

1. Que pensez-vous du terme « humanités numériques » et de l'utilisation de l'intelligence artificielle (IA) pour automatiser certaines tâches dans votre domaine d'étude ?
2. Dans quelle mesure est-il important de comprendre la signification culturelle et historique d'un projet comme le nôtre, et dans quelle mesure est-il important de transmettre cette signification dans le produit final ?

Questions sur les Stèles Bouddhistes

Représentation, Contexte Historique et Interprétation

1. Dans quelle mesure les stèles représentent-elles fidèlement la vie des personnes pour lesquelles elles ont été créées ?
2. Les stèles constituent-elles des représentations exactes de la vie des individus, ou ont-elles tendance à exagérer certains aspects de leur identité, de leurs réalisations ou de leur statut ?
3. Les informations contenues dans les stèles sont-elles suffisamment détaillées pour permettre une compréhension significative de la vie et des accomplissements d'un individu, ou servent-elles principalement de résumés succincts ?
4. Quelle est aujourd'hui l'importance de l'étude de l'histoire économique à travers les stèles ?

5. Dans quelle mesure l'histoire représentée dans les stèles diffère-t-elle des documents « officiels » produits par les gouvernements ou les autorités de la même période ?
6. Quelle importance la date de création d'une stèle a-t-elle dans l'interprétation de sa signification ? Les interprétations peuvent-elles changer considérablement selon la période historique ?
7. Quelle variation existe-t-il entre les stèles provenant de différentes régions ? Est-il possible de généraliser la signification des symboles lorsque ces derniers peuvent avoir eu des interprétations différentes selon les régions ou les périodes ?

Style Artistique, Symbolisme et Classification

1. Existe-t-il des styles artistiques régionaux distincts représentant un même animal ou motif qui devraient être considérés comme des classes séparées ?
2. Quels concepts bouddhistes sont les plus importants à comprendre lors de l'analyse des stèles (par exemple : karma, impermanence, cosmologie) ?
3. Comment la cosmologie bouddhiste influence-t-elle l'organisation spatiale et l'orientation des éléments présents sur les stèles ?
4. Existe-t-il des périodes de déstandardisation stylistique (comme les XVIIe ou XVIIIe siècles) durant lesquelles les conventions artistiques établies ont été volontairement modifiées ?
5. Comment pouvons-nous nous assurer qu'un modèle d'IA ne classe pas les variations stylistiques créatives ou les exceptions artistiques comme des erreurs ou comme des catégories distinctes ?
6. Dans les stèles bouddhistes, chaque animal ou objet représenté possède-t-il une signification symbolique, ou certains éléments sont-ils principalement décoratifs ?
7. Lorsque le dragon et le phénix apparaissent ensemble, leur association modifie-t-elle la signification de chacun par rapport à leur apparition séparée ? Cette relation devrait-elle être considérée comme une catégorie distincte ?
8. Comment les créatures hybrides (par exemple, un dragon se transformant en nuage) devraient-elles être classifiées ? Devraient-elles constituer des classes séparées ou être considérées comme des attributs liés à une catégorie principale ?

9. Lors de l'identification des éléments décoratifs, est-il nécessaire d'enregistrer leur position sur la stèle comme attribut obligatoire ?
10. De nombreux ornements semblent transmettre des récits historiques complexes liés au statut sacré, à la légitimité historique, au rang social et aux tendances artistiques. L'interprétation de ces récits peut-elle être confiée efficacement à une IA, ou l'IA devrait-elle principalement servir à organiser les ensembles de données tandis que l'interprétation reste du ressort des experts humains ?

Questions sur l'IA et l'Interprétation Historique

1. Lorsqu'elle dispose d'un contexte suffisant, l'IA peut-elle être utilisée pour interpréter correctement des données historiques ?
2. L'IA peut-elle produire de manière autonome des interprétations fiables de documents historiques et de matériaux patrimoniaux, ou l'expertise humaine reste-t-elle indispensable ?
3. Si l'IA pouvait théoriquement interpréter des documents et des artefacts historiques avec une précision égale ou supérieure à celle des experts humains, devrait-elle devenir la méthode privilégiée d'analyse ? Ou les chercheurs humains devraient-ils conserver l'autorité finale sur l'interprétation ?
4. Puisque les modèles d'IA peuvent contenir des biais inhérents, quels aspects de l'histoire des stèles bouddhistes, du symbolisme ou de la représentation culturelle risquent d'être sous-représentés ?

Questions sur Notre Projet et la Modélisation des Données

Représentation et Relations

1. Comment pouvons-nous garantir que les perspectives historiques moins visibles, marginalisées ou sous-représentées soient incluses dans notre ensemble de données ?
2. Comment les interprétations des stèles sont-elles utilisées aujourd'hui ? Sont-elles liées à des questions modernes telles que l'identité culturelle, les revendications historiques ou les conflits territoriaux ?
3. Comment les chercheurs déterminent-ils les relations entre les entités représentées dans les bases de données archéologiques et historiques ?

4. Lors de la segmentation des images de stèles, les modèles d'IA devraient-ils classifier des motifs isolés (par exemple, une seule griffe de dragon) ou des compositions complètes (par exemple, un dragon poursuivant une perle) ? Quelle unité d'analyse reste la plus stable à travers les siècles ?
5. Quels motifs, relations ou connexions symboliques courants présents dans les stèles devrions-nous connaître, et comment pouvons-nous réussir à les identifier ?

Fichiers d'Autorité et Entraînement de l'IA pour les Objets du Patrimoine Culturel

Métadonnées et Structure

1. Quelles sont les propriétés les plus importantes qui doivent être incluses dans un fichier d'autorité pour les objets archéologiques afin de le rendre utile à l'entraînement d'une IA ?
2. Comment gérer la variabilité terminologique (par exemple, noms régionaux, synonymes, termes historiques versus termes modernes) dans un fichier d'autorité contrôlé ?
3. Quel niveau de précision est approprié pour classifier les types d'objets afin d'obtenir un apprentissage automatique efficace (par exemple : « récipient » versus « amphore » versus « amphore à vin ») ?
4. Comment représenter les identifications incertaines ou partielles (par exemple : « possible fibule » ou « fragment d'objet inconnu ») sans perturber le modèle d'IA ?
5. Quelle est la meilleure pratique pour représenter les relations hiérarchiques entre catégories d'objets (par exemple : céramique → poterie → bol) dans un format lisible par machine ?
6. Comment gérer les variations temporelles et culturelles dans la classification des objets lorsque le même type d'objet peut avoir différents noms selon les périodes ou les régions ?
7. Quelles métadonnées devraient accompagner chaque terme d'autorité (par exemple : définitions, images, références bibliographiques, scores de confiance) afin d'améliorer la qualité de l'entraînement de l'IA ?
8. Quelles sont les erreurs ou difficultés les plus fréquentes que vous avez observées lors de la création de fichiers d'autorité pour des projets d'IA dans le domaine du patrimoine culturel ?

9. Si vous pouviez concevoir aujourd'hui une structure idéale de fichier d'autorité spécifiquement destinée à l'entraînement d'une IA, que modifieriez-vous par rapport aux normes existantes telles que le Getty Art & Architecture Thesaurus (AAT) ou Dublin Core ?

Appendix C: LLM Prompts Used for OCR

Prompt 1

This prompt was used to run OCR on the original scans of L'Art à Hué to produce a TXT file version of the French, Vietnamese, and Chinese characters.

“Run a multilingual (French, Vietnamese, and Chinese) OCR on this document and output the text exactly as written.”

Prompt 2

This prompt was used to translate the output of Prompt 1, which was a transcription of the original French, Vietnamese, and Chinese characters, to English. It is important to note that only the French text was translated. The rest of the transcription was left untouched because the Vietnamese and Chinese characters were part of the subject of the text with the intention of analyzing them using the context derived from the French explanations.

“Translate the French text below to English, but leave the Vietnamese and Chinese characters the same. The translation should retain the original text format, including italics, bold font, and paragraph separations such that it can be copy and pasted.”

Appendix D: Database Node Categories and Relationships

Categories

- Character
- Inanimate_Objects
 - Eight_Jewels
- Flora
 - Flower
 - Fruit
 - Leaf
- Animal
 - Non_Supernatural_Animal
 - * Bat
 - * Lion
 - * Fish
 - Supernatural_Animal
 - * Dragon
 - True_Dragon
 - Face_Dragon
 - Serpent_Dragon
 - * Phoenix

Relationships

- GROUP_OF
 - Source node: category of the group
 - Target node: the group node
 - The target node is a group of nodes in the source node category. This node is the target of the MEMBER_OF relationship.
- MEMBER_OF
 - Source node: a member of a group

- Target node: the group this member belongs to
- This node is part of a group with other members that all point to a target node. This target node is related to a category via the GROUP_OF relationship.
- RELATED_BY
 - Source node: one member of the relationship
 - Target node: the relating node
 - The source node and another source node in the dataset point to the same target node via the RELATED_BY relationship. These nodes are combined to create new nodes (see Appendix B).
- TRANSFORMS_INTO
 - Source node: the beginning object
 - Target node: the resulting object
 - Some entities transform into others. The source node transforms into the target node in this relationship.
- TYPE_OF
 - Source node: the parent node
 - Target node: the child node
 - This relationship is the most common within the database. The target node is the lower level of a hierarchy (the child node) of a higher level (the parent node).

Property Keys

- label : the default key; the English word describing the object
- en : the English translation of the label; according to Google Gemini
- fr : the French translation of the label from the original text
- vn : the Vietnamese translation of the label from the original text (does not apply to all nodes)
- zh : the Chinese translation of the label from the original text (does not apply to all nodes)

Appendix E: GitHub Repository File Structure

The GitHub repository at the link below contains all of the documentation, code, output, and other resources produced by our methods. <https://github.com/wpilyone26archeology/automation-of-methodologies-for-archival-processes.git>

The file structure is outlined below:

```
automation-of-methodologies-for-archival-processes
├── authority-file
│   ├── authorityFile.txt
│   ├── "Entity List.xlsx"
│   ├── method-for-creating-an-authority-file-database
│   └── Neo4j-access-codes.txt
├── document-preservation
│   ├── "Dublin Core.xmp"
│   ├── method-for-appending-metadata-to-files-for-document-preservation
│   ├── "L'Art à Hue.xmp"
│   └── RenameToDCIdentifier.jsx
├── text-analysis-ocr
│   └── method-for-performing-multilingual-ocr-and-transcribing-textual-documents
├── ai-pipeline-for-image-recognition-and-classification
├── README
└── stelemotifpipeline2.py
```

Appendix F: “Method for Appending Metadata to Files for Document Preservation”

Purpose

The purpose of this process is to append metadata to image files so that they contain all necessary information to enable automatic classification, simplified data organization, and future document reconstruction. The metadata follows the Dublin Core standard and the files should be saved in a lossless compression TIFF file format, as recommended by the Library of Congress (Dublin Core Metadata Initiative (DCMI), n.d.; Library of Congress, n.d.-b).

System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

Adobe Photoshop

Adobe Bridge

Excel

Toolkits

Metadata Deluxe toolkit

Custom Scripts

Custom Dublin Core metadata template

“Rename to DC Identifier (Fixed)” custom script

Methodology

Follow the instructions in the GitHub Repository section to ensure you have all of the required custom script downloads and extensions for appending the metadata.

SAVE THE IMAGES AS TIFF FILES

System Requirements:

- *Adobe Photoshop*

1. Open Adobe Photoshop.
2. Select File → Scripts → Image Processor.
 - a. In Section 1, select the folder that contains the images you wish to process.
 - b. In Section 2, you can select the location that you wish to save the processed images to. If this does not work, you can leave the location as the default and the processed images will save to a subfolder called TIFF in the original file location.
 - c. In Section 3, uncheck “Save as JPEG” and check “Save as TIFF.”
 - d. Click Run.

APPEND METADATA

System Requirements:

- *Text editor (e.g. Notepad, Visual Studio Code)*
- *Adobe Bridge*
- *Custom Dublin Core metadata template*
- *Metadata Deluxe toolkit*
- *Excel*

Creating the Metadata Template

1. *If this is your first time using the metadata template, please follow the Metadata Template Download instructions in Section F.*
2. Open the Dublin Core metadata template in your text editor (e.g. Notepad, Visual Studio Code).
3. Edit the fields with metadata that is consistent across the full batch of data. These are fields that are not specific to individual images (e.g. creator, source, etc.). For the full description of the metadata fields and examples of added data, see the table in Appendix G.

Appending the Metadata

Ensure that you have a working metadata template before completing this process.

1. Open Adobe Bridge.
2. In Adobe Bridge, navigate to the folder that contains the TIFF images. Use CTRL+A to select them all.
3. Select Tools → Append Metadata → Dublin Core. The application should automatically append all the available Metadata to the selected images.
4. Select Metadata Deluxe → Metadata Export-Import.
 - a. *If this is your first time using Metadata Deluxe, please follow the Metadata Deluxe Install instructions in Section F.*
 - b. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core
 - ii. ←Export→
 - iii. Files to Export : Entire Folder
 - iv. Options
 1. Prepend Image dates with ‘_’
 2. Remove line breaks
 - c. Select Export.
 - d. Save the file as a *.txt file in the folder with the processed images.
 - e. A success window should appear.
5. Open Excel.
6. Select Open → Browse → [navigate to the folder with the file].
 - a. Ensure that you change the file specification in the bottom right to “All Files”.
 - b. Select “Finish” (you don’t need to change anything on the Text Import Wizard screen).
7. The only blank columns in the spreadsheet should be Title, Subject, Description, Contributor, Identifier.
8. Add the individual image information to the associated rows in the respective columns. See Appendix G for an example.

9. Save and close Excel.
10. In Adobe Bridge, navigate to Metadata Deluxe → Metadata Export-Import.
 - a. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core
 - ii. →Import←
 - iii. Metadata text file : [name of the file]
 - iv. Match on : Filename
 - v. Existing Data : Write only to empty fields
 - b. Select Import.
11. To double check that the import was successful, follow step 4 to export the file again. You should not overwrite the existing file in case there was an issue. Once you have confirmed the export was successful, you can delete the original export.

CREATE A MASTER LIST

System Requirements:

- *Adobe Bridge*
- *Custom Dublin Core metadata template*
- *Metadata Deluxe extension*
- *Excel*
- *“Rename to DC Identifier (Fixed)” custom script*

1. Copy all the images from all the folders to a master folder.
2. *If this is your first time using the “Rename to DC Identifier (Fixed)” custom script, please follow the Custom Script Install instructions in Section F*
3. Open Adobe Bridge.
4. Navigate to the master folder and use CTRL+A to select all of the images in that folder.
5. Select Metadata Deluxe → Metadata Export-Import.
 - a. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core

- ii. ←Export→
 - iii. Files to Export : Entire Folder
 - iv. Options
 1. Prepend Image dates with ‘_’
 2. Remove line breaks
 - b. Select Export.
 - c. Save the file as a *.txt file in the folder with the processed images.
 - d. A success window should appear.
6. Open Excel.
7. Select Open → Browse → [navigate to the folder with the file].
 - a. Ensure that you change the file specification in the bottom right to “All Files”.
 - b. Select “Finish” (you don’t need to change anything on the Text Import Wizard screen).
8. Insert a column to the right of the File Name column.
 - a. Title this column “New File Name.”
 - b. Copy the Identifier column into this new column.
 - c. Save and close.
9. In Adobe Bridge, use CTRL+A to select all of the images in the master folder. Right click, then select “Rename to DC Identifier (Fixed)”.
 - a. Select the *.txt file that you just closed and run the script.
10. All of your files should now be labeled with their associated Dublin Core identifiers.

Installation Instructions

GITHUB REPOSITORY

The GitHub repository contains all of the necessary custom scripts required to follow this procedure.

1. Navigate to the GitHub repository [here](#).
2. Select the green “Code” button. Then select “Download ZIP.”
3. Navigate to your Downloads folder and extract the contents of the “automation-of-methodologies-for-archival-processes-main” folder.

METADATA TEMPLATE DOWNLOAD

The metadata template contains the XMP code base for the Dublin Core metadata standard. Use the instructions below to download this template and implement it in Adobe Bridge.

1. Download the supporting files using the GitHub Repository instructions in Section F.
2. Open File Explorer.
3. Select View → Hidden Items.
4. Navigate to automation-of-methodologies-for-archival-processes → document-preservation. Copy the Dublin Core file.
5. Navigate to Users → [Your User Account] → AppData → Roaming → Adobe → XMP → Metadata Templates.
6. Paste the copied metadata template file in this folder.

METADATA DELUXE INSTALL

Metadata Deluxe is “a JavaScript extension for Adobe Bridge (2015 and higher) exports and imports XMP metadata embedded in media files using a tab-delimited text file (spreadsheet)” (Metadata Deluxe, n.d.).

1. Navigate to [Metadata Deluxe](#).
2. Select the green Download button.

- a. A Javascript JSX file titled, `export_import_2023-08-16.jsx`, should download to your computer.
3. Open Adobe Bridge.
4. Navigate to Edit → Preferences → Startup Scripts → Reveal Scripts in Explorer.
 - a. File Explorer should automatically open to the scripts folder.
5. Move the downloaded file from your Downloads folder to the newly opened folder.
6. Restart Adobe Bridge.
 - a. On startup, a prompt should appear asking to enable the script. Select Yes to enable the script.
7. A new item should appear on the navbar at the top titled “Metadata Deluxe.” You can now use this extension to import and export metadata as *.txt files that can be opened and edited in Excel.

CUSTOM SCRIPT INSTALL

1. Open a text editor (e.g. VSCode).
2. If you have not already, download the supporting files using the GitHub Repository instructions Section J.
3. Navigate to `automation-of-methodologies-for-archival-processes` → `document-preservation`. Copy the `RenameToDCIdentifier` file.
4. Open Adobe Bridge.
5. Navigate to Edit → Preferences → Startup Scripts → Reveal Scripts in Explorer.
 - a. File Explorer should automatically open to the scripts folder.
6. Paste the copied file in this folder.
7. Close all programs and restart Adobe Bridge.
 - a. On startup, a prompt should appear asking to enable the script. Select Yes to enable the script.

8. When you right click an image in Adobe Bridge, a new item should appear at the bottom of the options titled “Rename to DC Identifier (Fixed).” You can now use this extension to rename files to their associated identifier included in the metadata.

Appendix G: Outline of Metadata Field Contents

This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (DCMI Usage Board, n.d.; Adobe, n.d.-d). Any fields in the Example Data column that begin with a * were not specified in the metadata template and were added to each image individually.

Table 1: This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (DCMI Usage Board, n.d.; Adobe, n.d.-d).

Property	Value Type	Description	Example Data
dc:title	Lang Alt	The title of the document, or the name given to the resource. Typically, it will be a name by which the resource is formally known.	*Object Name
dc:creator	seq PropertyName	The authors of the resource (listed in order of precedence, if significant).	Léopold Michel Cadière, et al.
dc:subject	bag Text	An unordered array of descriptive phrases or keywords that specify the topic of the content of the resource.	*From the Authority File
dc:description	Lang Alt	A textual description of the content of the resource. Multiple values may be present for different languages.	*From the Image Captions
dc:publisher	bag ProperName	Contributors to the resource (other than the authors).	

Table 1: This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (DCMI Usage Board, n.d.; Adobe, n.d.-d).

Property	Value Type	Description	Example Data
dc:date	seq Date	Date(s) that something interesting happened to the resource.	1919
dc:type	bag open Choice	A document type; for example, novel, poem, or working paper.	StillImage, Text, ImageFixe, Texte
dc:format	MIMEType	The file format used when saving the resource. Tools and applications should set this property to the save format of the data. It may include appropriate qualifiers.	image/tiff
dc:identifier	Text	Unique identifier of the resource.	HUE-ch#-p#-desc
dc:source	Text	Unique identifier of the work from which this resource was derived.	<i>L'Art à Hué</i>
dc:language	bag Locale	An unordered array specifying the languages used in the resource.	fr, vi, zh
dc:relation	bag Text	Relationships to other documents	https://revues.vietnamica.online/bavh/19_1bavh/

Table 1: This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (DCMI Usage Board, n.d.; Adobe, n.d.-d).

Property	Value Type	Description	Example Data
dc:coverage	Text	The extent or scope of the resource.	19th-Century Vietnamese Artwork
dc:rights	Lang Alt	Informal rights statement, selected by language.	Public Domain, Public Domain, Domaine Public

Information about *L'Art à Hué* was included from the Paris Museums website (Paris Musées, n.d.-a).

Appendix H: Unique Field Metadata Outline

1. Title : Object Name in Human Text (e.g. Planche 110)
2. Description : Image Caption or Text Summary in “English; French” format
 - a. Example: Accent ornaments; Ornaments d’accent
3. Identifier : HUE-ch#-p#-title
 - a. Example: HUE-ch4-p181-planche110
4. Subject: keywords from the Authority File that describe the contents of the page in “English (object 1); French (object 1); Vietnamese (object 1); Chinese (object 1); English (object 2); French (object 2)...” format
 - a. Example: “animal; animaux; dragon; con rồng; 龍”
5. Contributors should be left blank.

Appendix I: “Method for Performing Multilingual OCR and Transcribing Textual Documents”

Purpose

The goal of this methodology is to process images (e.g. binarization, fix orientation, content selection, DPI settings, etc.) and then perform multilingual object character recognition (OCR) capable of recognizing multiple languages in order to output machine-readable PDF files that can be translated into other languages.

System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

ScanTailor

Methodology

*Note: **Bold** typeface indicates project-specific values. These fields can vary depending on the scan settings and the text itself.*

SCANTAILOR

System Requirements:

- *ScanTailor*

1. *If this is your first time using ScanTailor, please follow the ScanTailor Install instructions in Section I.*
2. Open a new project in ScanTailor.
3. Fix the DPI of all pages to **400 x 400**.
 - a. In the “Fix DPI” window (“Tools” → “Fix DPI...”), select the “All Pages” tab.

- b. Click “All Pages” in the smaller window. Once you have made your selection, the options at the bottom of the window should be available for you to change.
 - c. In the dropdown, select “**400 x 400**”.
 - d. Click apply.
4. Remove any duplicate pages from the project.
5. Fix orientation of all pages such that they are in a portrait layout.
6. Set “Split Pages” to “Split Manually” on “All Pages” so that the software doesn’t automatically split the pages in the next step.
7. Run “Deskew” on the “Automatic” setting on “All Pages”. Fix manually as needed such that text lines are parallel to on-screen gridlines.
8. Use “Select Content” on “All Pages” to highlight all text, images, page numbers, and annotations on each page. Manually adjust as needed.
9. Use the “Margins” tab to add the margins to each page, omitting the scanner background.
 - a. Decouple the “Top” and “Bottom,” “Left” and “Right” fields.
 - b. Adjust the margins in each field until the entire page is within the selected region.
10. In the “Output” tab, do the following:
 - a. Apply an “Output Resolution” of **400dpi** to “All Pages”.
 - b. Apply “Black and White” to “All Pages”.
 - c. Leave all other settings at their default.
 - d. Run.
 - e. *Note: ScanTailor keeps the file names, but loses the metadata information. The metadata will need to be re-appended to the exported files.*
11. “File” → “Save Project As...” to save the project file and output in an appropriate folder (usually the input directory).
12. The output files are located in the input directory in a new folder labeled “OUT”.

GOOGLE GEMINI

System Requirements:

- *Internet access and access to Google Gemini*

Optical Character Recognition

1. Open a new chat in Google Gemini.
2. Upload one ScanTailor TIFF output file and copy the following prompt:
 - a. “Run a multilingual (**French, Vietnamese, and Chinese**) OCR on this document and output the text exactly as written.”
 - b. The **bold** text above should be changed depending on the languages included in your document.
3. Copy and paste the output into a text file (e.g. Word Document, Google Doc, .txt file).
 - a. Make sure that the text is readable, keeping the italics and other special font formats.
 - b. Correct dashes as necessary (these used to be words that broke over the line).
4. Continue this process for all of the text pages in the chapter.

English Transcription

1. Open a new chat in Google Gemini.
2. Upload one ScanTailor TIFF output file and copy the following prompt:
 - a. “Translate the **French** text below to **English**, but leave the **Vietnamese and Chinese characters the same**. The translation should retain the original text format, including italics, bold font, and paragraph separations such that it can be copy and pasted.”
 - b. The **bold** text above should be changed depending on the languages included in your document.
3. Copy and paste the output into a text file (e.g. Word Document, Google Doc, .txt file).
 - a. Make sure that the text is readable, keeping the italics and other special font formats.
 - b. Correct dashes as necessary (these used to be words that broke over the line).
4. Continue this process for all of the text pages in the chapter.

Installation Instructions

SCANTAILOR INSTALL

“ScanTailor is an interactive post-processing tool for scanned pages – page splitting, deskewing, borders, and more” (ScanTailor, n.d.).

1. Navigate to [ScanTailor](#).
2. Select the blue “Downloads & binaries” button. You should now be on the Downloads page. Navigate to the “Github project page” using the link on the website.
3. On GitHub, scroll down to the README file. Below the “Installation and Tips” header, select the “installer” link.
4. Below the “Installation and first start” header, follow the necessary instructions to install and start the application.

Appendix J: “Method for Creating an Authority File Database”

Purpose

The purpose of this process is to create a reference framework (authority file) that contains entities and relationships between them. This framework should be in the form of a graphical database and should contain necessary translations and alternate terminology for all entities.

System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

Neo4j

Spreadsheet software (e.g. Google Sheets)

Methodology

Follow the instructions in the GitHub Repository section to ensure you have all of the required custom script downloads and extensions for appending the metadata.

OUTLINE THE NODES AND RELATIONSHIPS

System Requirements:

- *Google Sheets*

1. Open Google Sheets.
2. Create a table with columns for all of the entity keys you wish to include and one column for the relations. See Appendix K for an example.
3. Read each of the text pages and list the entities referenced in the text as individual rows. See Appendix K for an example.

For a full example of the spreadsheet, see the Entity Table file in the GitHub repository at [automation-of-methodologies-for-archival-processes](#) → [authority-file](#).

USE CYPHER TO CREATE THE DATABASE

System Requirements:

- Text editor (e.g. Notepad, Visual Studio Code)

1. Open a text editor (e.g. Notepad, Visual Studio Code).
2. Define all of the entities in the Google Sheet using MERGE statements like the one below.

```
MERGE (temp_id:Category {  
    property0: 'Property 0',  
    property1: 'Property 1',  
    property2: 'Property 2'  
})
```

- a. temp_id – the temporary id for the node used in this query.
 - b. Category – the category of the node (e.g. Dragon).
 - c. property# – the name of a property key of the node (e.g. label).
 - d. Property # – the information associated with the property# property key (e.g. Dragon).
3. Define all of the relationships in the Google Sheet using MERGE statements like the one below.

```
MERGE (temp_id0-[:RELATION]->(temp_id1))
```

- a. temp_id0 – the temporary id for the source node used in this query.
- b. temp_id1 – the temporary id for the target node used in this query.
- c. :RELATION – the defined relation between the two nodes (e.g. :TYPE_OF).

For a full example of Cypher code, see the authorityFile.txt file in automation-of-methodologies-for-archival-processes → authority-file. The [Cypher Manual](#) contains more information about MERGE statements and other commands you can use in Neo4j (Neo4j, Inc., n.d.-b).

RUN QUERIES IN NEO4J

System Requirements:

- *Neo4j*

1. Open [Neo4j](#)
 - a. Neo4j is a graphical database management and organization software (Neo4j, Inc., 2026).
 - b. *Create an account if you do not already have one.*
2. *If this is your first time using Neo4j,*
 - a. *Create an account.*
 - b. *Create an instance. All Neo4j databases run on instances.*
 - i. *Select the Instances tab on the left, then select the blue “Create instance” button.*
 - ii. *Select the Free option, then name your instance and fill out the remaining information. It will take a few minutes for the software to finish creating the instance.*
3. Once your instance appears on the “Instances” page, select Open → Query.
4. Copy the Cypher code from your text editor and paste it into the text box at the top (“Start typing...”).
5. Select the blue run button. Your database should now be complete.

Installation Instructions

GITHUB REPOSITORY

The GitHub repository contains all of the necessary custom scripts required to follow this procedure.

1. Navigate to the GitHub repository [here](#).
2. Select the green “Code” button. Then select “Download ZIP.”
3. Navigate to your Downloads folder and extract the contents of the “automation-of-methodologies-for-archival-processes-main” folder.

Interactive Example

This example is the result of the Cypher code and entity list included in the shared GitHub repository. Follow the instructions below to view a completed database on Neo4j.

1. Navigate to workspace.neo4j.io.
2. Select "Connect to an Instance."
3. Enter the following information.
 - a. Connection URL: `neo4j+s://13a7c7f8.databases.neo4j.io`
 - b. Database User: `13a7c7f8`
 - c. Password: `__BCM1nPsk2SLlaNwLZyov7FJm6QC9JkVOkmopT5tHY`
4. Select "Database Overview". From here, you can see information about the database nodes, relationships, and property keys.
 - a. Nodes
 - i. The number next to the "Nodes" header indicates the number of nodes in the database.
 - ii. Each of the items below the header represents a different category of object.
 - iii. Double click on any of the items to see all the nodes in that category. The * will show all of the nodes. You may need to edit the provided code and remove "LIMIT #" to see the full database.
 - b. Relationships
 - i. The number next to the "Relationships" header indicates the total number of relationships in the database.
 - ii. Each of the items below the header represents a different type of relationship.
 - iii. Double click on any of the items to see all the relationships of that type. The * will show all of the nodes. You may need to edit the provided code and remove "LIMIT #" to see the full database.
 - c. Property Keys
 - i. The items below the header represent the different keys associated with each node.

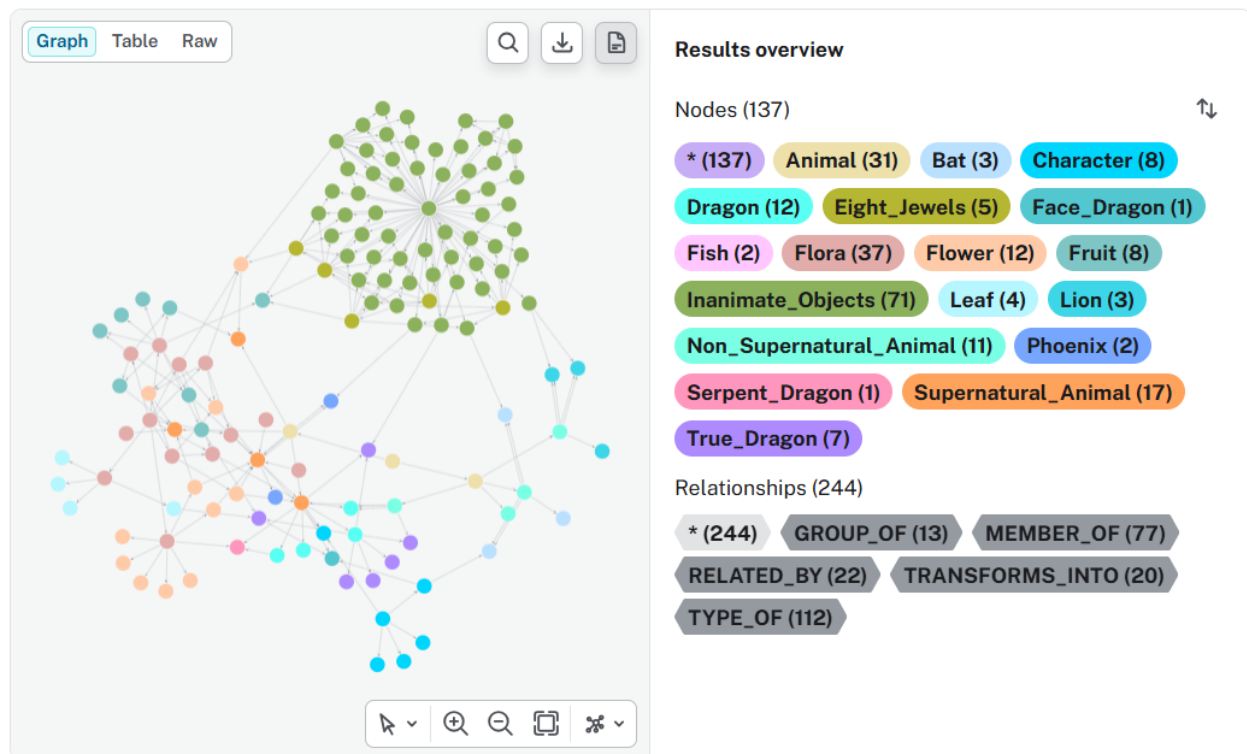


Figure 16: A screenshot of the full database created from the entities and relationships in chapters two through five of *L'Art à Hué*. The breakdown of the categories and relationships defined in the database is listed on the right.

Appendix K: Entities and Relationships in *L'Art à Hué*

Columns in the Google Sheet

1. Entity Name (EN)
2. Entity Name (FR)
3. Entity Name (VN)
4. Entity Name (ZH)
5. Category
6. Relations

Entities and Relationships

1. Every entity should have an English (EN) and French (FR) translation.
2. Some will have a Vietnamese (VN) and/or a Chinese (ZH) translation.
3. The category refers to the chapter the entity was referenced in (e.g. Characters, Inanimate Objects, Flowers, etc.). The categories included in the *L'Art à Hué* authority file were:
 - a. Character
 - b. Inanimate Objects
 - i. Eight Jewels
 - c. Flora
 - i. Fruit
 - ii. Flower
 - iii. Leaf
 - d. Animal
 - i. Supernatural Animal
 1. Dragon
 - a. True Dragon
 1. Face Dragon
 - b. Serpent Dragon
 2. Pheonix
 - ii. Non-Supernatural Animal

1. Bat
2. Lion
3. Fish
4. The relations to other objects should only be listed if they are part of any of the following five categories:
 - a. The entity is a group of other objects.
 - i. Example: The Flaming Globe.
 - b. The entity is a member of a group of objects.
 - i. Example: The Disc of the Moon, the Clouds, and the Flames are part of The Flaming Globe.
 - c. The entity is a type of another entity.
 - i. Example: Dragon is a Supernatural Animal.
 - d. The entity transforms into another entity.
 - i. Example: the Ribbons transform into the Flames.
 - e. The entity is related to another entity via a third entity.
 - i. Example: the Dragon and the Fish are related by The Fish Changing into a Dragon.

Appendix L: "An AI Pipeline for Image Recognition and Classification"

System Requirements

Please note that these instructions are intended for a Windows operating system.

- **CUDA toolkit** - for SAM2 GPU acceleration if using NVIDIA GPU.
- **Python 3.10+**
- **LM Studio** - to locally run Qwen3
- **Conda** - for managing the gsam2 environment

Python Environment

To be installed in gsam2 conda environment.

- **Torch + Torchvision** - PyTorch, required by SAM2
- **Sam2** - SAM2 package, installed from GitHub repo
- **OpenAI** - Python client used to talk to LM Studio's local server
- **Pillow** - used for image loading and manipulation
- **Numpy** - for array operations
- **Matplotlib** - generated overlay visualizations
- **Tifffile** - for loading .tif files

Model Files

- SAM2 checkpoint, sam2.1_hiera_large.pt, was used due to accuracy.
 - Small/tiny variants exist, will be quicker but less accurate
 - Installation into checkpoints folder is recommended within the same root directory as pipeline code
 - > root directory
 - > checkpoints folder
 - > sam2.1_hiera_large.pt
 - > stelemotifpipeline2.py
- **SAM2 config** - comes with cloned SAM2 repo
- **Qwen model of choice** - installed and ran within LM Studio

Code Files

- **stelemotifpipeline2.py** - Custom / edited pipeline combining Qwen3-VL-8B running locally + SAM2 running locally.
- **reference_library.py** - reference crop management and few-shot prompt builder.
- **test_connection.py** - diagnostic only, confirms LM Studio is running and reachable.

Note: stelemotifpipeline2.py and reference_library must be in the same directory.

Input and Output

Input Files

- Input files should in a recognizable location, depicted within pipeline code.
 - Altering stele scans to inverted black and white .tif files is recommended for optimal detection.
 - * White on black background may be marginally advantageous for object detection models compared to colored images.
 - * Gets reverted for final product.
 - Original colored .jpg file of stele should also be provided for segmentation and final output with bounding boxes and masks.

Output

- Specify a directory for outputs of the pipeline.
 - Final .jpg of stele with bounding boxes and masks overlaid, separate masks and .json file.

Directory Structure

```
AI_Image_detection_Pipeline/
├── stelemotifpipeline2.py
├── reference_library.py
├── test_connection.py
├── checkpoints/
│   └── sam2.1_hiera_large.pt
├── configs/
│   └── sam2.1/
│       └── sam2.1_hiera_l.yaml
├── sam2/ cloned SAM2 repo, installed as package
├── test_images/
│   ├── 10925bwi.tif
│   └── 10925.jpg
├── output/ generated automatically on first run
│   ├── detection_overlay.png
│   ├── detections.json
│   ├── raw_detections.json
│   └── mask_000_dragon.png
```